INCH-POUND

MIL-M-28787/352
31 October 1989

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

68000 CONTROL/MAINTENANCE MODULE, KEY CODE ADP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic 68000 control maintenance module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 5962

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbook The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

DOD-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.5513 failures per million operating hours when predicted in accordance with MIL-HDBK-217E.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be ADP.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 120 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Pin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	S	-------Module shall pass all remaining tests----------						
D1-D20	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D21	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	mA
D22,D23	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D24-D26	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D27	Ibd	V	S	0.0	1.4	0.0	1.4	0.0	1.4	mA
D28-D30	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D31	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D32	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D33-D41	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D42	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D43-D51	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D52	Ibd	V	S	0.0	0.15	0.0	0.15	0.0	0.15	mA
D53	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D54-D56	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D57	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D58	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D59	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	mA
E1-E20	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E21-E27	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E28-E30	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E31	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E32-E53	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E54-E57	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E58,E59	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V

- See footnotes at end of table

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TABLE I Module electrical requirements - Continued.

Test and symbol	Parameter 5	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
F1-F4	Ios	VII	S	-130	-15	-130	-15	-130	-15	mA
F5-F10	Ios	VII	S	-130	-30	-130	-30	-130	-30	mA
F11-F13	Ios	VII	S	-100	-40	-100	-40	-100	-40	mA
F14-F21	Ios	VII	S	-100	-15	-100	-15	-100	-15	mA
F22-F23	Ios	VII	S	-110	-15	-110	-15	-110	-15	mA
F24	Ios	VII	S	-100	-40	-100	-40	-100	-40	mA
F25	Ios	VII	S	-100	-15	-100	-15	-100	-15	mA
F26-F33	Ios	VII	S	-130	-15	-130	-15	-130	-15	mA
F34	Ios	VII	S	-100	-15	100	-15	-100	-15	mA
G1-G20	Iih	VIII	X	0.0	20.0	0 0	20.0	0.0	20.0	uA
G21	Iih	VIII	X	0.0	115.0	0 0	115.0	0.0	115.0	uA
G22.G23	Iih	VIII	X	0.0	75.0	0.0	75.0	0 0	75.0	uA
G24-G26	Iih	VIII	X	0.0	50.0	0 0	50.0	0.0	50.0	uA
G27	Iih	VIII	X	0.0	130 0	0 0	130.0	0 0	130.0	uA
G28 G30	Iih	VIII	X	0.0	25.0	0 0	25.0	0.0	25 0	uA
G31	Iih	VIII	X	0 0	75 0	0 0	75 0	0 0	75 0	uA
G32	Iih	VIII	X	0.0	60.0	0 0	60.0	0.0	60.0	uA
G33 G41	Iih	VIII	X	0.0	20.0	0 0	20 0	0.0	20.0	uA
G42	Iih	VIII	X	0.0	100.0	0 0	100.0	0.0	100.0	uA
G43 G51	Iih	VIII	X	0.0	20.0	0 0	20 0	0.0	20.0	uA
G52	Iih	VIII	X	-1.08	-0.773	-1 08	-0.773	-1.08	-0.773	mA
G53	Iih	VIII	X	0.0	20 0	0 0	20.0	0.0	20.0	uA
G54 G56	Iih	VIII	X	0.0	50.0	0 0	50.0	0.0	50.0	uA
G57	Iih	VIII	X	0 0	75 0	0 0	75 0	0 0	75 0	uA
G58	Iih	VIII	X	0.0	40 0	0.0	40.0	0.0	40.0	uA
G59	Iih	VIII	X	0.0	100.0	0 0	100.0	0.0	100.0	uA
H1-H20	Iil	IX	X	-440.0	0.0	-440 0	0 0	-440 0	0.0	uA
H21	Iil	IX	X	-3.17	-1 07	-3 17	-1 07	-3 17	-1 07	mA
H22 H23	Iil	IX	X	-2.25	-1.0	-2 25	-1 0	-2 25	-1.0	mA
H24 H26	Iil	IX	X	-2.0	-1 0	-2.0	-1.0	2 0	-1.0	mA
H27	Iil	IX	X	-2.76	-1.06	-2 76	-1.06	-2 76	-1.06	mA
H28-H30	Iil	IX	X	-250.0	-10.0	-250 0	-10.0	-250 0	-10 0	uA
H31	Iil	IX	X	-2.29	-1.03	-2.29	-1.03	-2 29	-1.03	mA
H32	Iil	IX	X	-800.0	-60.0	-800 0	-60.0	-800 0	-60.0	uA

- See footnotes at end of table

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H33	Iil	IX	X	-400.0	-30.0	-400.0	-30.0	-400 0	-30.0	uA
H34-H41	Iil	IX	X	-400.0	-100.0	-400.0	-100.0	-400 0	-100.0	uA
H42	Iil	IX	X	-2.4	-0.35	-2 4	-0 35	-2.4	-0 35	mA
H43,H44	Iil	IX	X	-400.0	-30 0	-400 0	-30 0	-400 0	-30 0	uA
H45-H51	Iil	IX	X	-380.0	-30 0	-380.0	-30 0	-380 0	-30.0	uA
H52	Iil	IX	X	-2.36	-1.80	-2.36	-1.80	-2 36	-1.80	mA
H53	Iil	IX	X	-400.0	-30 0	-400.0	-30 0	-400 0	-30.0	uA
H54-H56	Iil	IX	X	-2 0	-1.0	-2 0	-1 0	-2 0	-1.0	mA
H57	Iil	IX	X	-2.29	-1 03	-2.29	-1.03	-2 29	-1.03	mA
H58	Iil	IX	X	-880.0	0.0	-880 0	0 0	-880 0	0.0	uA
H59	Iil	IX	X	-2 16	0 0	-2 16	0.0	-2 16	0.0	mA
J1-J10	Voh	X	X	2.4	4.5	2 4	4.5	2.4	4.5	V
J11-J25	Voh	X	X	2.5	4.5	2 5	4.5	2.5	4.5	V
J26-J33	Voh	X	X	2 4	4.5	2 4	4.5	2.4	4.5	V
J34	Voh	X	X	2.5	4.5	2 5	4 5	2.5	4.5	V
J35	Voh	X	X	2.4	2.6	2 4	2.6	2.4	2 6	V
K1-K4	Vol	XI	X	0 0	0.4	0 0	0 4	0.0	0 4	V
K5-K10	Vol	XI	X	0 0	0.5	0 0	0 5	0.0	0.5	V
K11	Vol	XI	X	0 2	0 5	0 2	0 5	0 2	0 5	V
K12,K13	Vol	XI	X	0.0	0.5	0 0	0 5	0.0	0.5	V
K14-K23	Vol	XI	X	0.0	0.4	0 0	0 4	0.0	0.4	V
K24	Vol	XI	X	0.2	0.5	0 2	0.5	0.2	0.5	V
K25-K34	Vol	XI	X	0 0	0.4	0 0	0.4	0.0	0.4	V
L1-L8	Iozh	XII	X	-20	20	-20	20	-20	20	uA
M1-M8	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
N	Icc	XIV	X	34	340	34	340	34	340	mA
P1,P2	Functional	XV	X	----------		See table XV				
Q1,Q2	Dynamic	XVI	S	----------		See table XVI				
R	PSFC	XVIII	S	3 9	7.3	3 9	7.3	3.9	7 3	uF

- See footnotes at end of table

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
T1-T35	Output crosstalk high level	XX	Q	0	400	0	400	0	400	mV
U1-U5	Output crosstalk low level	XXI	Q	0	400	0	400	0	400	mV
U6-U22	Output crosstalk low level	XXI	Q	0	500	0	500	0	500	mV
U23-U26	Output crosstalk low level	XXI	Q	0	600	0	600	0	600	mV
U27-U34	Output crosstalk low level	XXI	Q	0	750	0	750	0	750	mV
V1-V4	Output crosstalk duration low level	XXII	Q	0	50	0	50	0	50	nS
V5-V12	Output crosstalk duration low level	XXII	Q	0	57	0	57	0	57	nS
V13-V20	Output crosstalk duration low level	XXII	Q	0	50	0	50	0	50	nS
W1-W3	Cin	XXIII	Q	2	20	2	20	2	20	pF
W4	Cin	XXIII	Q	2	25	2	25	2	25	pF
W5-W10	Cin	XXIII	Q	2	20	2	20	2	20	pF

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
W11	Cin	XXIII	Q	2	25	2	25	2	25	pF
W12-W20	Cin	XXIII	Q	2	20	2	20	2	20	pF
W21	Cin	XXIII	Q	2	25	2	25	2	25	pF
W22	Cin	XXIII	Q	2	35	2	35	2	35	pF
W23	Cin	XXIII	Q	2	40	2	40	2	40	pF
W24	Cin	XXIII	Q	6	50	6	50	6	50	pF
W25,W26	Cin	XXIII	Q	4	30	4	30	4	30	pF
W27	Cin	XXIII	Q	4	35	4	35	4	35	pF
W28	Cin	XXIII	Q	4	40	4	40	4	40	pF
W29	Cin	XXIII	Q	4	35	4	35	4	35	pF
W30	Cin	XXIII	Q	2	30	2	30	2	30	pF
W31,W32	Cin	XXIII	Q	2	35	2	35	2	35	pF
W33	Cin	XXIII	Q	2	30	2	30	2	30	pF
W34,W35	Cin	XXIII	Q	2	25	2	25	2	25	pF
W36	Cin	XXIII	Q	4	50	4	50	4	50	pF
W37	Cin	XXIII	Q	6	40	6	40	6	40	pF
W38	Cin	XXIII	Q	2	15	2	15	2	15	pF
W39,W40	Cin	XXIII	Q	2	20	2	20	2	20	pF
W41-W43	Cin	XXIII	Q	2	15	2	15	2	15	pF
W44,W45	Cin	XXIII	Q	2	20	2	20	2	20	pF
W46-W48	Cin	XXIII	Q	2	15	2	15	2	15	pF
W49-W59	Cin	XXIII	Q	2	20	2	20	2	20	pF
X1-X8	Cout	XXIV	Q	2	20	2	20	2	20	pF
Y1-Y8	Disable time (tplz)	XXV	Q	5	45	5	45	5	45	nS
Y9 Y16	Disable time (tphz)	XXV	Q	5	66	5	66	5	66	nS
Z	Threshold	XXVI	X	See table XXVI						

- See footnotes on next page

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/352. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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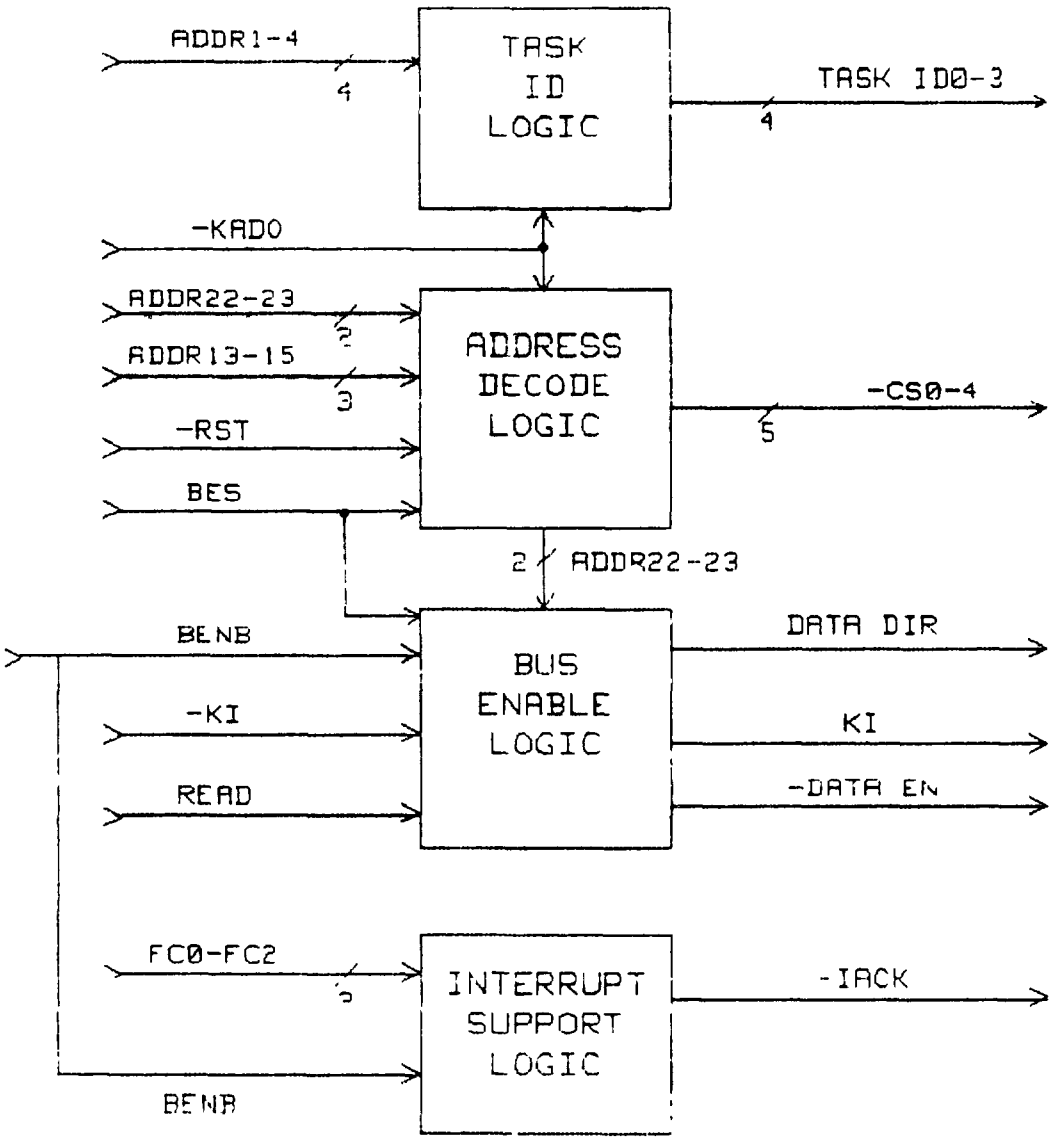


FIGURE 1. ADP functional block diagram.

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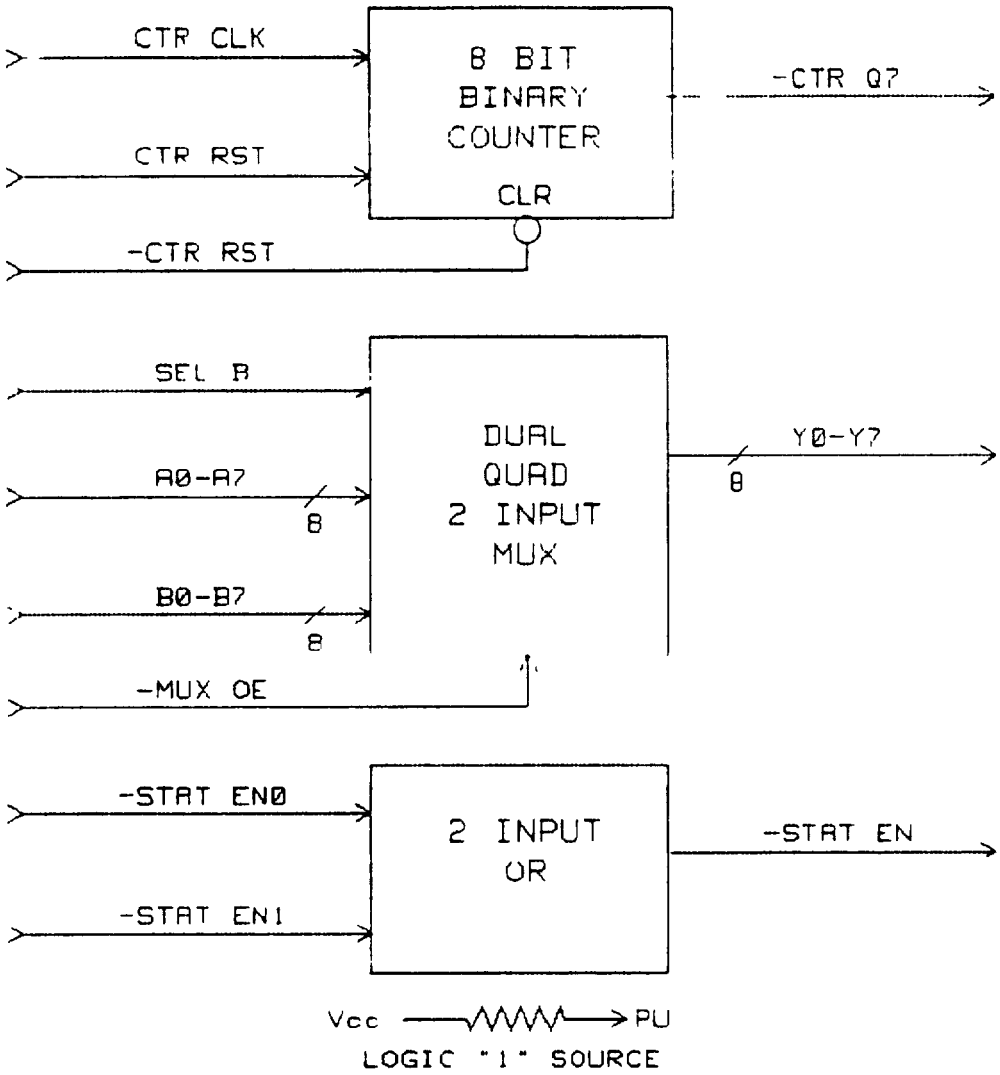


FIGURE 1. ADP functional block diagram - Continued.

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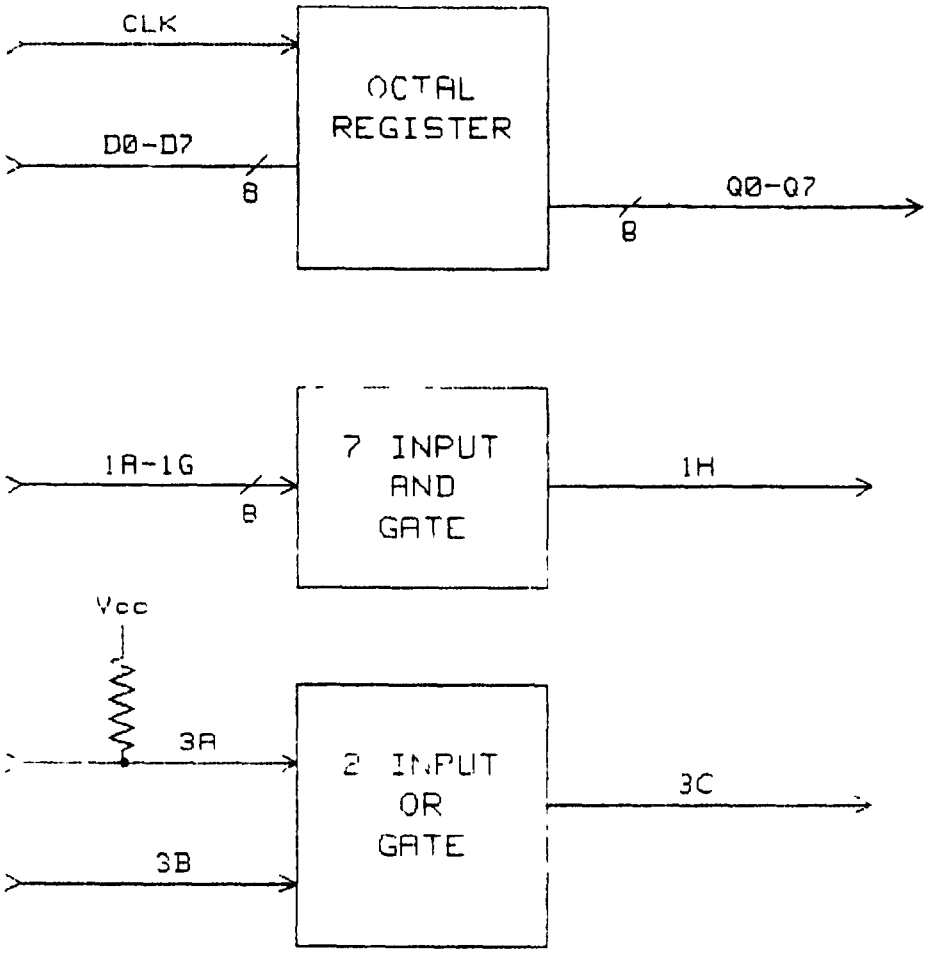


FIGURE 1. ADP functional block diagram - Continued.

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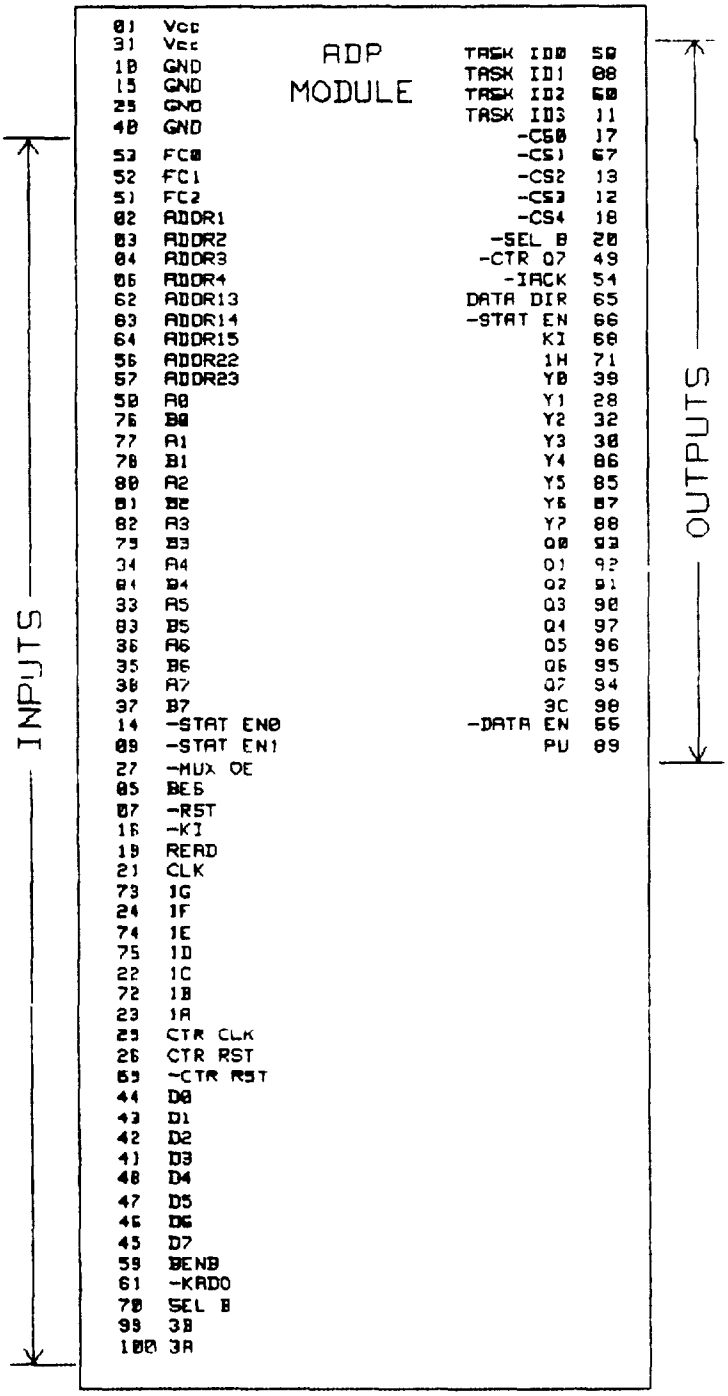


FIGURE 2. ADP pin assignment.

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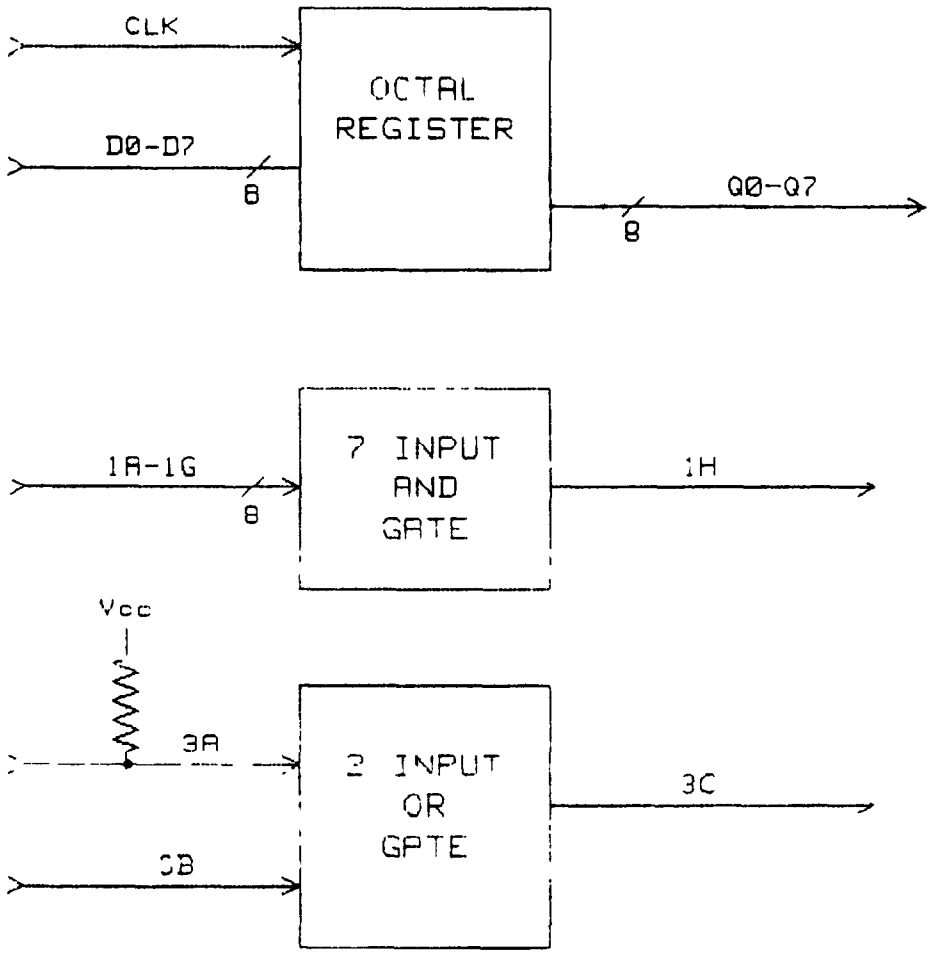


FIGURE 1. ADP functional block diagram - Continued.

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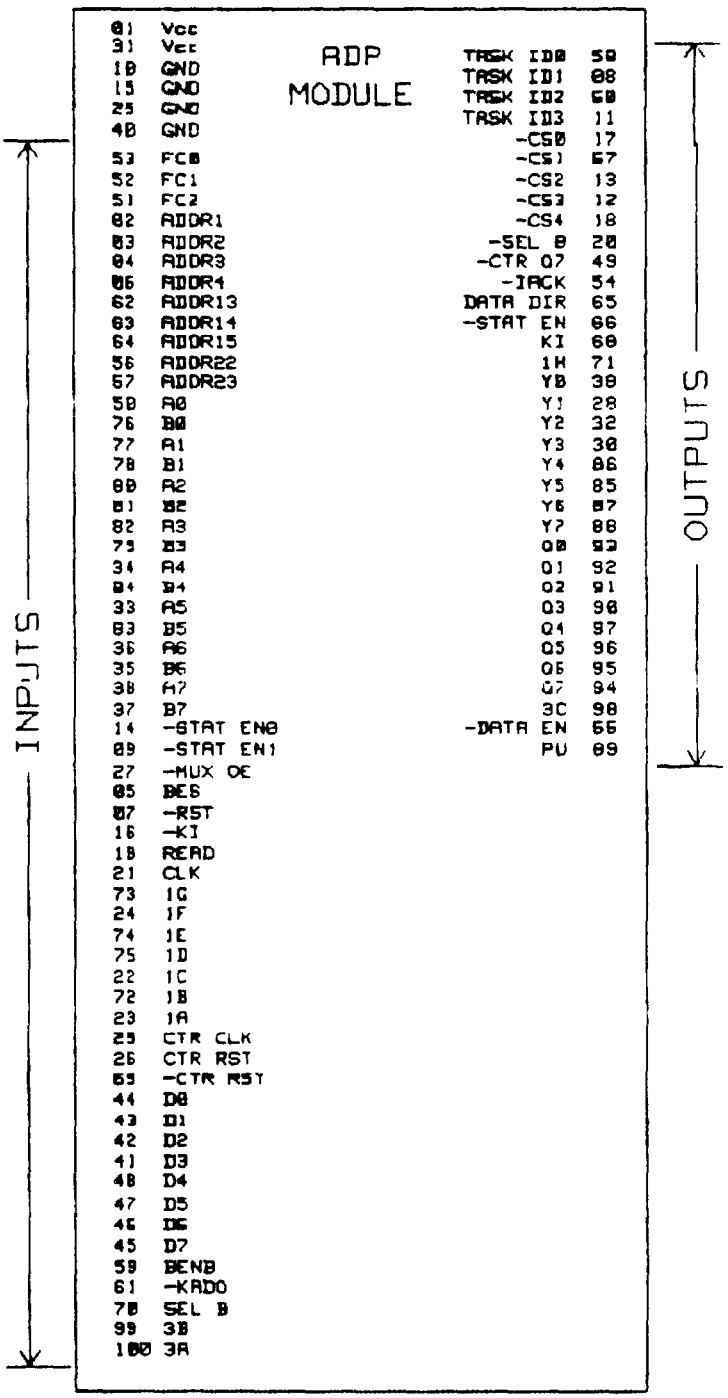


FIGURE 2. ADP pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. Test conditions A and C through F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

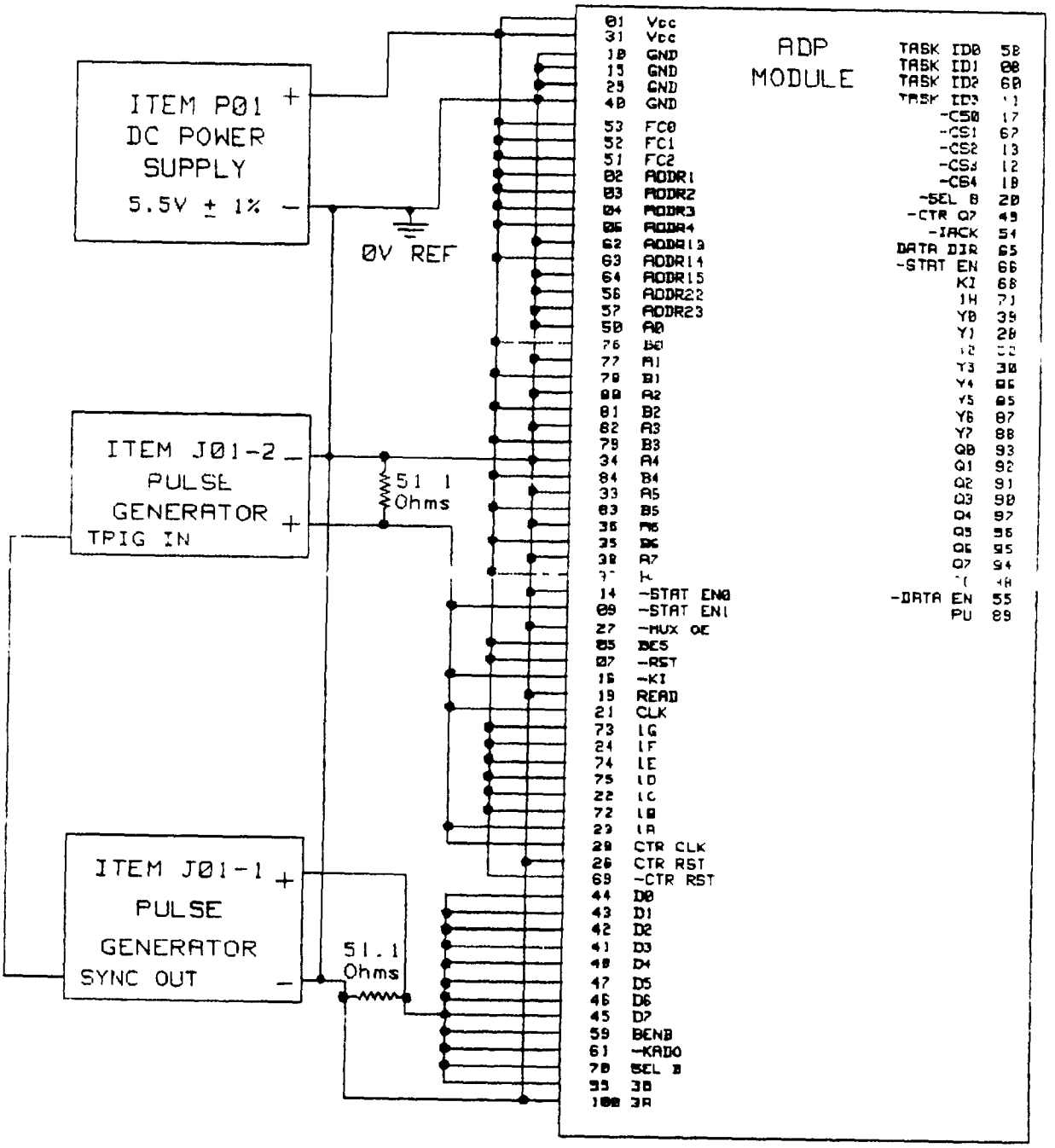
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3 and shall not violate the timing requirements shown on figure 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 5 and 6.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 7. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. For load information, see output voltage, high level, test.

FIGURE 3. ADP operating temperature and life test circuit.

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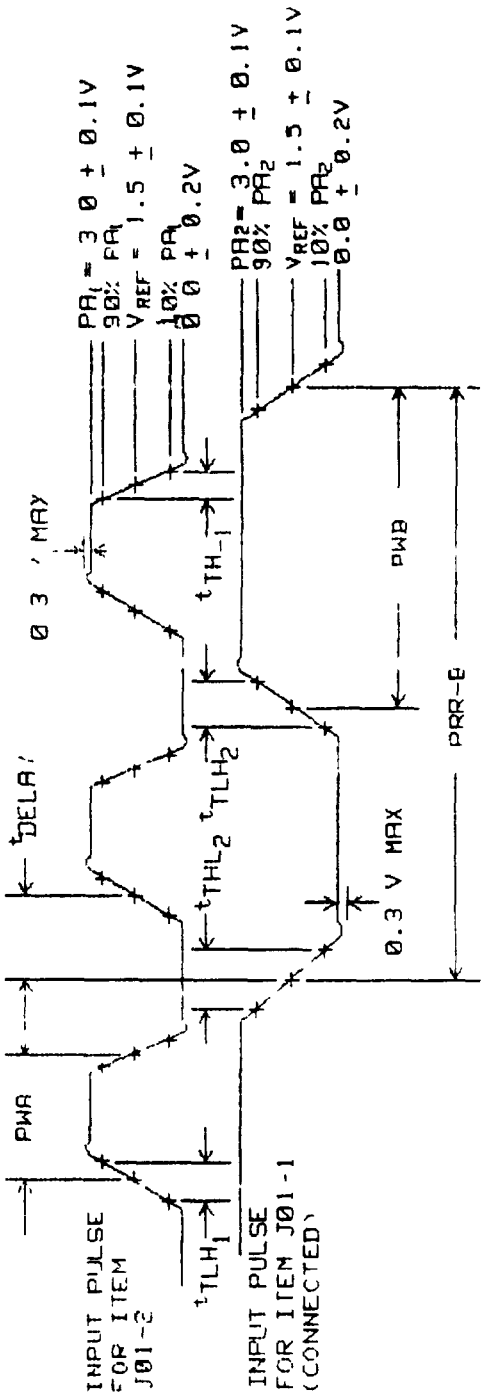


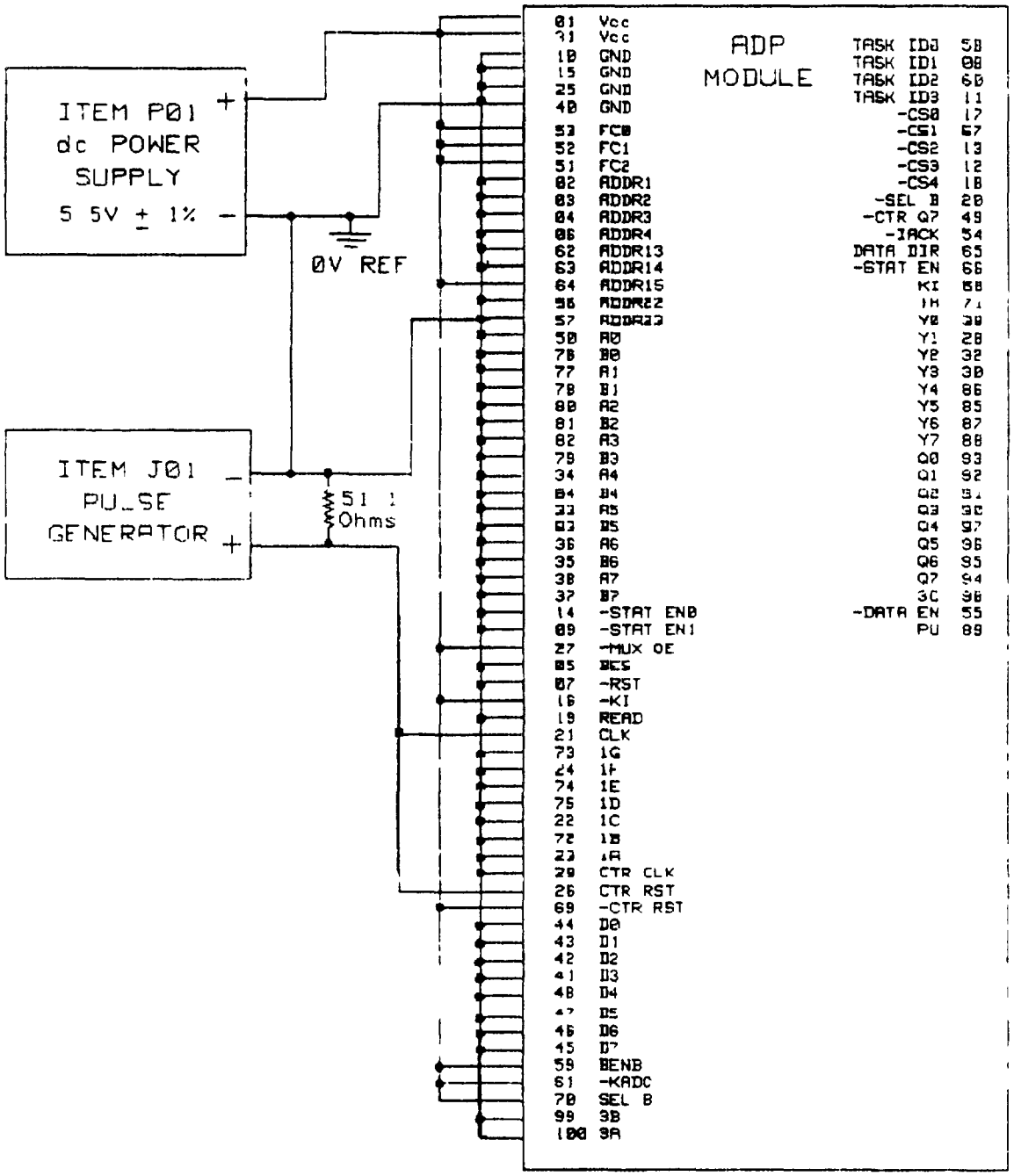
FIGURE 4. Operating temperature and life test waveforms.

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NOTES:

1. Input pulse parameters for item J01-2.
   (a) PWA = 250 ± 25 nS.
   (b) tTLH1 = tTHL1 = 50 ± 30 nS.
   (c) Double pulse mode.
   (d) tDELAY = 150 ± 50 nS.
2. Input pulse parameters for item J01-1.
   (a) PWB = 500 ± 50 nS.
   (b) tTLH2 = tTHL2 = 50 ± 30 nS.
   (c) PRR-B = 1000 ± 100 nS.
3. All voltages are referenced to ground.

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NOTE:

1. For load information, see output voltage, high level, test.

FIGURE 5. ADP component temperature test circuit.

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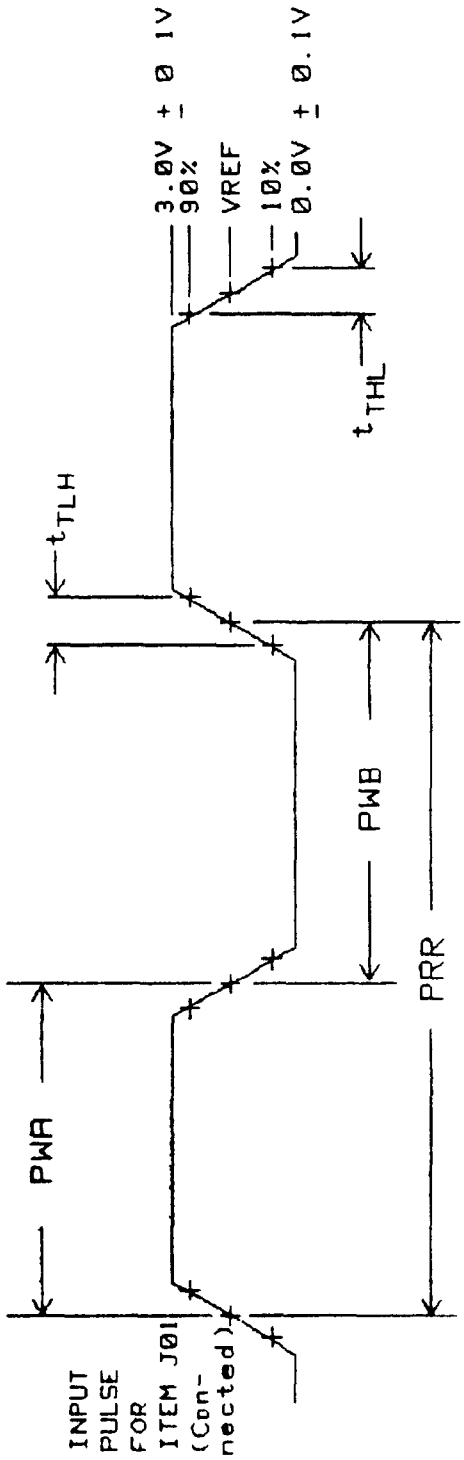


FIGURE 6. Component temperature test waveforms.

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NOTES:

1. Input pulse parameters for item J01.

   (a) PWA = 250 ± 25 nS.

   (b) PWB = 250 ± 25 nS.

   (c) PRR = 500 ± 50 nS.

   (d) tTLH = tTHL = 50 ± 10 nS.

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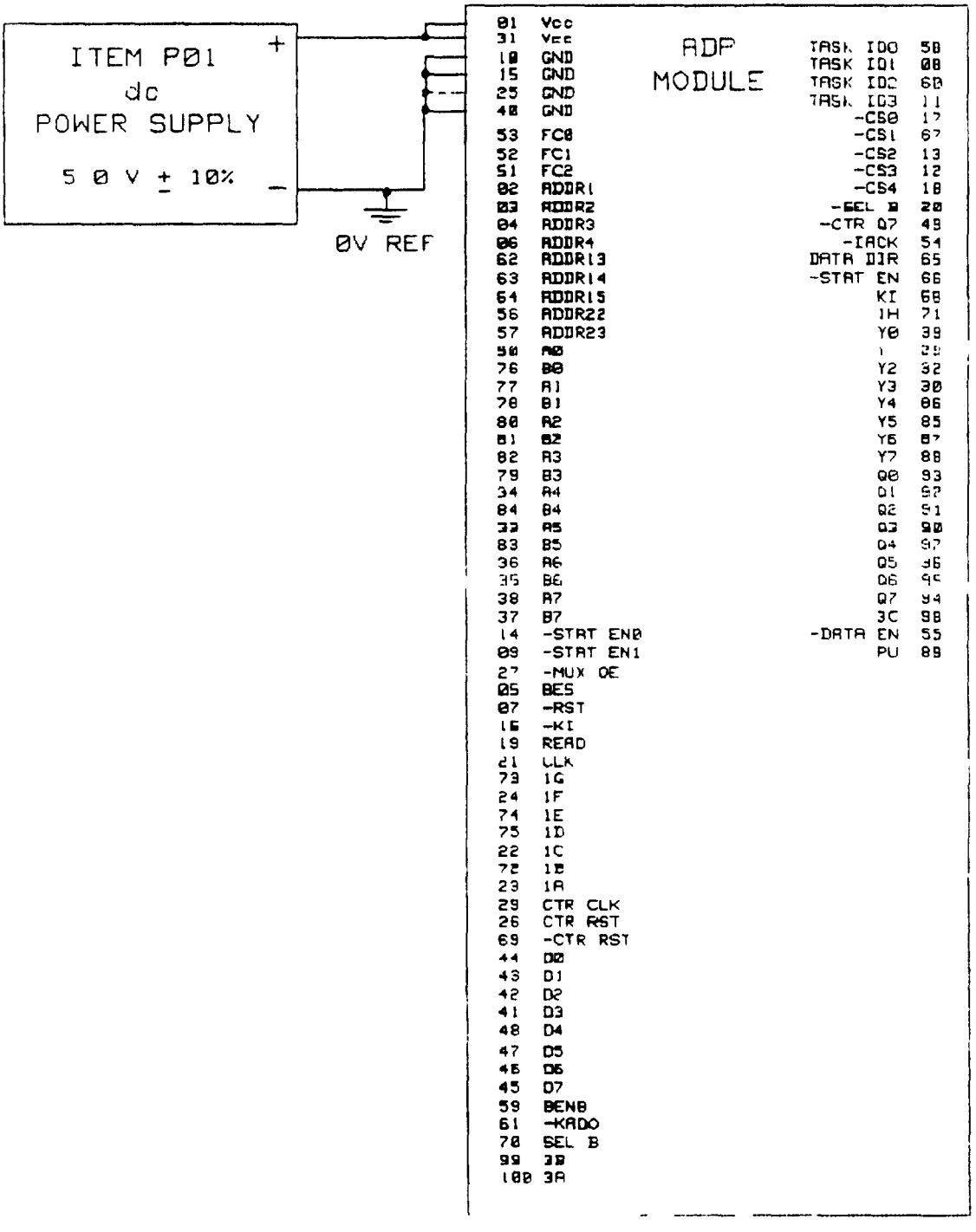


FIGURE 7. ADP vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, C through V, Y and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through Y and Z shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 1.868 W (max); 1.016 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
ADDR1	-1.5	0.7	2.0	-0.44	20	100	20
ADDR2	-1.5	0.7	2.0	-0.44	20	100	20
ADDR3	-1.5	0.7	2.0	-0.44	20	100	20
ADDR4	-1.5	0.7	2.0	-0.44	20	100	25
A0	-1.5	0.7	2.0	-0.44	20	100	35
A1	-1.5	0.7	2.0	-0.44	20	100	20
A2	-1.5	0.7	2.0	-0.44	20	100	20
A3	-1.5	0.7	2.0	-0.44	20	100	20
A4	-1.5	0.7	2.0	-0.44	20	100	20
A5	-1.5	0.7	2.0	-0.44	20	100	20
A6	-1.5	0.7	2.0	-0.44	20	100	20
A7	-1.5	0.7	2.0	-0.44	20	100	20
B0	-1.5	0.7	2.0	-0.44	20	100	20

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
B1	-1.5	0.7	2.0	-0.44	20	100	20
B2	-1.5	0.7	2.0	-0.44	20	100	20
B3	-1.5	0.7	2.0	-0.44	20	100	20
B4	-1.5	0.7	2.0	-0.44	20	100	20
B5	-1.5	0.7	2.0	-0.44	20	100	20
B6	-1.5	0.7	2.0	-0.44	20	100	20
B7	-1.5	0.7	2.0	-0.44	20	100	20
-KADO	-1.2	0.7	2.0	-3.17	115	2200	50
ADDR22	-1.2	0.8	2 0	-2.25	75	2000	30
ADDR23	-1.2	0.8	2.0	-2.25	75	2000	30
ADDR13	-1.2	0.8	2.0	-2.0	50	1000	20
ADDR14	-1.2	0.8	2.0	-2.0	50	1000	20
ADDR15	-1.2	0.8	2.0	-2.0	50	1000	20

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
BES	-1.2	0.7	2.0	-2.76	130	1400	35
-STAT EN 1	-1.5	0.8	2.0	-0.25	25	1000	30
-STAT EN 0	-1.5	0.8	2.0	-0.25	25	1000	40
READ	-1.5	0.8	2.0	-0.25	25	1000	25
-KI	-1.2	0.8	2.0	-2.29	75	2000	40
-RST	-1.5	0.7	2.0	-0.80	60	300	30
CLK	-1.5	0.7	2.0	-0.40	20	100	35
D0	-1.5	0.7	2.0	-0.40	20	100	20
D1	-1.5	0.7	2.0	-0.40	20	100	15
D2	-1.5	0.7	2.0	-0.40	20	100	15
D3	-1.5	0.7	2.0	-0.40	20	100	15
D4	-1.5	0.7	2.0	-0.40	20	100	15
D5	-1.5	0.7	2.0	-0.40	20	100	15

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D6	-1.5	0.7	2.0	-0.40	20	100	15
D7	-1.5	0.7	2.0	-0.40	20	100	20
CTR CLK	-1.5	0.7	2.0	-2.4	100	400	25
-CTR RST	-1.5	0.7	2.0	-0.40	20	100	20
CTR RST	-1.5	0.7	2.0	-0.40	20	100	35
1A	-1.5	0.7	2.0	-0.38	20	100	20
1B	-1.5	0.7	2.0	-0.38	20	100	20
1C	-1.5	0.7	2.0	-0.38	20	100	15
1D	-1.5	0.7	2.0	-0.38	20	100	20
1E	-1.5	0.7	2.0	-0.38	20	100	20
1F	-1.5	0.7	2.0	-0.38	20	100	20
1G	-1.5	0.7	2.0	-0.38	20	100	25
3A	-1.5	0.7	2.0	-2.22	-1000	100	25

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
3B	-1.5	0.7	2.0	-0.40	20	100	15
FCO	-1.2	0.8	2.0	-2.0	50	1000	20
FC1	-1.2	0.8	2.0	-2.0	50	1000	20
FC2	-1.2	0.8	2.0	-2.0	50	1000	20
BENB	-1.2	0.8	2.0	-2.29	75	2000	50
-MUX OE	-1.5	0.7	2.0	-0.88	40	200	35
SEL B	-1.5	0.7	2.0	-2.16	100	500	40

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)	Cout max (pF)
TASK ID0	2.4	0.4	-1.0	12.0	NA	NA	-15	NA
TASK ID1	2.4	0.4	-1.0	12.0	NA	NA	-15	NA
TASK ID2	2.4	0.4	-1.0	12.0	NA	NA	-15	NA
TASK ID3	2.4	0.4	-1.0	12.0	NA	NA	-15	NA
-CS0	2.4	0.5	-2.0	8.0	NA	NA	-30	NA
-CS1	2.4	0.5	-2.0	8.0	NA	NA	-30	NA
-CS4	2.4	0.5	-2.0	8.0	NA	NA	-30	NA
KI	2.4	0.5	-2.0	8.0	NA	NA	-30	NA
DATA DIR	2.4	0.5	-2.0	8.0	NA	NA	-30	NA
-STAT EN	2.4	0.5	-2.0	8.0	NA	NA	-30	NA
-DATA EN	2.5	0.5	-1.0	20.0	NA	NA	-40	NA
-CS2	2.5	0.5	-1.0	20.0	NA	NA	-40	NA
-CS3	2.5	0.5	-1.0	20.0	NA	NA	-40	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)	Cout max (pF)
Q0	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
Q1	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
Q2	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
Q3	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
Q4	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
Q5	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
Q6	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
Q7	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
1H	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
3C	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
-IACK	2.5	0.5	-1.0	20.0	NA	NA	-40	NA
-CTR Q7	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
-SEL B	2.5	0.4	-0.4	4.0	NA	NA	-15	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)	Cout max (pF)
Y0	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
Y1	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
Y2	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
Y3	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
Y4	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
Y5	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
Y6	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
Y7	2.4	0.4	-1.0	12.0	20.0	20.0	-15	25.0
PU	2.4	NA	-2.0	NA	NA	NA	NA	NA

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6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test and the disable time test.

6.4 Module function. The module function is not printed herein because of its length. If required, a copy may be obtained from Naval Weapons Support Center (Code 602), Crane, IN 47522-5060.

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6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Functional description
ADDR1 ADDR2 ADDR3 ADDR4	2 3 4 6	I I I I	Inputs to a multiplexer which are selected when -KADO is inactive (high). Corresponding outputs are TASK ID0-TASK ID3 (respectively).
A0 A1 A2 A3 A4 A5 A6 A7	50 77 80 82 34 33 36 38	I I I I I I I I	Data inputs to an octal, two input multiplexer. These inputs are selected when SEL B is low.
B0 B1 B2 B3 B4 B5 B6 B7	76 78 81 79 84 83 35 37	I I I I I I I I	Data inputs to an octal, two input multiplexer. These inputs are selected when SEL B is low.
-KADO	61	I	An active low control signal that overrides all address decode on the module. When -KADO is inactive (high), inputs ADDR1-ADDR4 correspond to outputs TASK ID0-TASK ID3, and ADDR13, ADDR14, ADDR15, ADDR22, and ADDR23 are decoded to generate chip selects. When -KADO is active (low), TASK ID0-TASK ID3 are low and -CS4 becomes active. All other chip selects are disabled, regardless of address inputs.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
ADDR23 ADDR22	57 56	I I	Address lines used to define address space that is outside the kernel. When ADDR22 or ADDR23 are high, all chip selects are disabled (except -CS4 may still be selected by driving -KADO low).
ADDR13 ADDR14 ADDR15	62 63 64	I I I	Three address lines that are decoded to generate -CS0 through -CS4. The module function (see 6.4) contains additional addressing information.
BES	5	I	An active high input which is intended to be active when address and data are valid. When BES and -KI are both high, -DATA EN becomes active. The falling edge of BES is used to clock in the -CS0 restore signal when ADDR13-ADDR15 are high and ADDR22 and ADDR23 are low.
-STAT EN0 -STAT EN1	14 9	I I	Inputs to a two input OR gate When -STAT EN0 and -STAT EN1 are low, -STAT EN is low.
READ	19	I	An input used to control DATA DIR. When READ is low and BENB is high, DATA DIR is low DATA DIR is also low when READ is high and ADDR22 and ADDR23 are low.
-KI	16	I	An active low input which is used to generate KI and -DATA EN. -DATA EN is active when -KI is high and BES is high. -KI is inverted to generate KI

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-RST	7	I	An active low input used to disable -CS0 after a processor reset. This allows the reset vectors to be located in the memory selected by -CS1.
CLK	21	I	A rising edge sensitive input which is used to latch data that appears on inputs D0-D7.
D0	44	I	Data inputs to an octal D-type flip-flop which are latched on the rising edge of the clock (CLK). Outputs Q0-Q7 correspond to inputs D0-D7.
D1	43	I	
D2	42	I	
D3	41	I	
D4	48	I	
D5	47	I	
D6	46	I	
D7	45	I	
CTR CLK	29	I	A falling edge sensitive input to a divide-by-256 counter. The output (-CTR Q7) changes state every 128 clock cycles unless interrupted by a clear input (CTR RST or -CTR RST).
CTR RST	26	I	An active high reset input to the divide-by-256 counter. Each time CTR RST goes high, the output (-CTR Q7) is set high and the counter is cleared.
-CTR RST	69	I	An active low reset input to the divide-by-256 counter. Each time -CTR RST goes low, the output (-CTR Q7) is set high and the counter is cleared.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
1A 1B 1C 1D 1E 1F 1G	23 72 22 75 74 24 73	I I I I I I I	Inputs to a seven input AND gate. When all seven inputs are high, the output (1H) is high.
3A 3B	100 99	I I	Inputs to a two input OR gate. When both inputs are low, the output (3C) is low. Input 3A is pulled up internally by a 2.74 KOhm resistor.
FC0 FC1 FC2	53 52 51	I I I	Three inputs to a four input NAND gate. When FC0-FC2 and BENB are all high, the output (-IACK) is low.
BENB	59	I	An active high input which is used to generate DATA DIR and -IACK. DATA DIR is low when BENB is high and READ is low. -IACK is low when FC0-FC2 and BENB are all high.
-MUX OE	27	I	An active low output control signal for the octal, two input multiplexer. When -MUX OE is low, outputs Y0-Y7 are enabled. When -MUX OE is high, Y0-Y7 are in the high impedence state.
SEL B	70	I	Select input to the octal, two input multiplexer. When SEL B is high, inputs B0-B7 are selected. When SEL B is low, inputs A0-A7 are selected. SEL B is also inverted to generate -SEL B.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
TASK ID0 TASK ID1 TASK ID2 TASK ID3	58 8 60 11	O O 	Outputs from a multiplexer that correspond to inputs ADDR1-ADDR4 (respectively) when -KADO is inactive (high). When -KADO is low, all outputs are low.
-CS0	17	O	An active low chip select output which is active when ADDR13, ADDR14, ADDR15, ADDR22 and ADDR23 are low, -KADO is high, and no reset has occurred. When -KADO is active, this chip select is disabled regardless of the address inputs. Each time -RST [illegible], -CS0 is restored by the falling edge of BES when ADDR13-ADDR15 are high and ADDR22 and ADDR23 are low.
-CS1	67	O	An active low chip select output which is active when ADDR14, ADDR15, ADDR22 and ADDR23 are low and ADDR13 and -KADO are high. When -KADO is active, this chip select is disabled. Each time -RST (reset) becomes active, -CS1 is also active for the address space normally associated with -CS0 until -CS0 is restored. The 68000 reset vectors must be located in the first eight locations of the address space normally selected by -CS1.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-CS4	18	O	An active low chip select output that is active when ADDR13, ADDR14, ADDR22 and ADDR23 are low and ADDR15 is high. When -KADO is active, this output is low regardless of the address inputs.
KI	68	O	An output that is generated by inverting -KI (an input).
DATA DIR	65	O	An output which is low when READ is low and BENB is high or READ is high and ADDR22 and ADDR23 are low.
-STAT EN	66	O	Output from a two input OR gate. When both inputs (-STAT EN0 and -STAT EN1) are low, -STAT EN is low.
-DATA EN	55	O	An active low output which is active when -KI and BES are high.
-CS2	13	O	An active low chip select output which is active when ADDR13, ADDR15, ADDR22 and ADDR23 are low and -KADO and ADDR14 are high This chip select is disabled when -KADO is active.
-CS3	12	O	An active low chip select output which is active when ADDR15, ADDR22 and ADDR23 are low and ADDR13, ADDR14, and -KADO are high. This chip select is disabled when -KADO is active.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7	93 92 91 90 97 96 95 94	O O O O O O O O	Outputs from an octal latch that correspond to D0-D7 (respectively). Data is latched on the rising edge of the clock input (CLK).
1H	71	O	Output of a seven input AND gate. When all seven inputs (1A-1G) are high, the output is high.
3C	98	O	Output from a two input OR gate. When both inputs (2A and 2B) are low, 3C is low.
-IACK	54	O	An output from a four input NAND gate. When FC0, FC1, FC2 and BENB are all low, -IACK is low.
-CTR Q7	49	O	Output of a divide-by-256 counter. The counter increments on the falling edge of each clock cycle (CTR CLK). When the counter reaches 128, -CTR Q7 goes low. When a reset input (-CTR RST or CTR RST) becomes active, the counter is reset.
Y0 Y1 Y2 Y3 Y4 Y5 Y6 Y7	39 28 32 30 86 85 87 88	O O O O O O O O	Outputs of an octal, two input multiplexer. Y0-Y7 correspond to A0-A7 (respectively) when the select input (SEL B) is low, and to B0-B7 (respectively) when the select input is high. These outputs are in the high-impedence state when -MUX OE is high and enabled when -MUX OE is low.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-SEL B	20	O	Output of an inverter whose input is SEL B.
PU	89	O	Logic "1" source for pulling up unused pins.

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 8.

(j) Vol test - table XI, figure 8.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 9.

(o) Dynamic test - tables XVI and XVII, figures 10 and 11.

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(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 12.

(r) Output crosstalk tests - tables XX and XXI, figures 13 and 14.

(s) Output crosstalk duration test - table XXII, figure 15.

(t) Cin test - table XXIII.

(u) Cout test - table XXIV.

(v) Disable time test - table XXV, figures 16 and 17.

(w) Threshold test - table XXVI.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0052)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40	0.0	1.0	Ohm	1-4
B5	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B6	Continuity, GND	10	15				
B7	Continuity, GND	15	25				
B8	Continuity, GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV, -70 mV	C
Power supply current limit	0.51 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	ADDR1	2	5.5	0.0	0.1	mA	1-3
D2	ADDR2	3					
D3	ADDR3	4					
D4	ADDR4	6					
D5	A0	50					
D6	A1	77					
D7	A2	80					
D8	A3	82					
D9	A4	34					
D10	A5	33					
D11	A6	36					
D12	A7	38					
D13	B0	76					
D14	B1	78					
D15	B2	81					
D16	B3	79					
D17	B4	84					
D18	B5	83					
D19	B6	35					
D20	B7	37	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	-KADO	61	5.5	0.0	2.2	mA	1-3
D22	ADDR23	57	5.5	0.0	2.0	mA	1-3
D23	ADDR22	56	5.5	0.0	2.0	mA	1-3
D24	ADDR13	62	5.5	0.0	1.0	mA	1-3
D25	ADDR14	63					
D26	ADDR15	64	5.5	0.0	1.0	mA	1-3
D27	BES	5	5.5	0.0	1.4	mA	1-3
D28	-STAT EN0	14	5.5	0.0	1.0	mA	1-3
D29	-STAT EN1	9					
D30	READ	19	5.5	0.0	1.0	mA	1-3
D31	-KI	16	5.5	0.0	2.0	mA	1-3
D32	-RST	7	5.5	0.0	0.3	mA	1-3
D33	CLK	21	5.5	0.0	0.1	mA	1-3
D34	D0	44					
D35	D1	43					
D36	D2	42					
D37	D3	41					
D38	D4	48					
D39	D5	47					
D40	D6	46	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	D7	45	5.5	0.0	0.1	mA	1-3
D42	CTR CLK	29	5.5	0.0	0.4	mA	1-3
D43	CTR RST	26	5.5	0.0	0.1	mA	1-3
D44	-CTR RST	69					
D45	1A	23					
D46	1B	72					
D47	1C	22					
D48	1D	75					
D49	1E	74					
D50	1F	24					
D51	1G	73	5.5	0.0	0.1	mA	1-3
D52	3A	100	5.5	0.0	0.15	mA	1-3
D53	3B	99	5.5	0.0	0.1	mA	1-3
D54	FC0	53	5.5	0.0	1.0	mA	1-3
D55	FC1	52					
D56	FC2	51	5.5	0.0	1.0	mA	1-3
D57	BENB	59	5.5	0.0	2.0	mA	1-3
D58	-MUX OE	27	5.5	0.0	0.2	mA	1-3
D59	SEL B	70	5.5	0.0	0.5	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D59
Power supply current limit	0.51 A	NA	D1-D59
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D51, D53-D59
	5.5 V	+ 55,-0 mV	D52
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D59
low, (VL)	0.0 V	± 100 mV	D1-D59
Input current measurement	NA	± 5.0 uA	D1-D20
	NA	± 110.0 uA	D21
	NA	± 100.0 uA	D22,D23
	NA	± 50.0 uA	D24-D26
	NA	± 70.0 uA	D27
	NA	± 50.0 uA	D28-D30
	NA	± 100.0 uA	D31
	NA	± 15.0 uA	D32
	NA	± 5.0 uA	D33-D41
	NA	± 20.0 uA	D42
	NA	± 5.0 uA	D43-D51
	NA	± 7.5 uA	D52
	NA	± 5.0 uA	D53
	NA	± 50.0 uA	D54-D56
	NA	± 100.0 uA	D57
	NA	± 10.0 uA	D58
	NA	± 25.0 uA	D59

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	ADDR1	2	-18.0	-1.5	-0.4	V	1-3
E2	ADDR2	3					
E3	ADDR3	4					
E4	ADDR4	6					
E5	A0	50					
E6	A1	77					
E7	A2	80					
E8	A3	82					
E9	A4	34					
E10	A5	33					
E11	A6	36					
E12	A7	38					
E13	B0	76					
E14	B1	78					
E15	B2	81					
E16	B3	79					
E17	B4	84					
E18	B5	83					
E19	B6	35					
E20	B7	37	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	-KADO	61	-18.0	-1.2	-0.4	V	1-3
E22	ADDR23	57					
E23	ADDR22	56					
E24	ADDR13	62					
E25	ADDR14	63					
E26	ADDR15	64					
E27	BES	5	-18.0	-1.2	-0.4	V	1-3
E28	-STAT EN0	14	-18.0	-1.5	-0.4	V	1-3
E29	-STAT EN1	9					
E30	READ	19	-18.0	-1.5	-0.4	V	1-3
E31	-KI	16	-18.0	-1.2	-0.4	V	1-3
E32	-RST	7	-18.0	-1.5	-0.4	V	1-3
E33	CLK	21					
E34	D0	44					
E35	D1	43					
E36	D2	42					
E37	D3	41					
E38	D4	48					
E39	D5	47					
E40	D6	46	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	D7	45	-18.0	-1.5	-0.4	V	1-3
E42	CTR CLK	29					
E43	CTR RST	26					
E44	-CTR RST	69					
E45	1A	23					
E46	1B	72					
E47	1C	22					
E48	1D	75					
E49	1E	74					
E50	1F	24					
E51	1G	73					
E52	3A	100					
E53	3B	99	-18.0	-1.5	-0.4	V	1-3
E54	FC0	53	-18.0	-1.2	-0.4	V	1-3
E55	FC1	52					
E56	FC2	51					
E57	BENB	59	-18.0	-1.2	-0.4	V	1-3
E58	-MUX OE	27	-18.0	-1.5	-0.4	V	1-3
E59	SEL B	70	-18.0	-1.5	-0.4	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E59
Power supply current limit	0.51 A	NA	E1-E59
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E59
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E59 E1-E59
Input voltage measurement	NA NA NA NA NA NA NA	± 55 mV ± 40 mV ± 55 mV ± 40 mV ± 55 mV ± 40 mV ± 55 mV	E1-E20 E21-E27 E28-E30 E31 E32-E53 E54-E57 E58,E59

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	TASK ID0	58	0.0	-130	-15	mA	1-4
F2	TASK ID1	8					
F3	TASK ID2	60					
F4	TASK ID3	11	0.0	-130	-15	mA	1-4
F5	-CS0	17	0.0	-130	-30	mA	1-4
F6	-CS1	67					
F7	-CS4	18					
F8	KI	68					
F9	DATA DIR	65					
F10	-STAT EN	66	0.0	-130	-30	mA	1-4
F11	-DATA EN	55	0.0	-100	-40	mA	1-4
F12	-CS2	13					
F13	-CS3	12	0.0	-100	-40	mA	1-4
F14	Q0	93	0.0	-100	-15	mA	1-4
F15	Q1	92					
F16	Q2	91					
F17	Q3	90					
F18	Q4	97					
F19	Q5	96					
F20	Q6	95	0.0	-100	-15	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	Q7	94	0.0	-100	-15	mA	1-4
F22	1H	71	0.0	-110	-15	mA	1-4
F23	3C	98	0.0	-110	-15	mA	1-4
F24	-IACK	54	0.0	-100	-40	mA	1-4
F25	-CTR Q7	49	0.0	-100	-15	mA	1-4
F26	Y0	39	0.0	-130	-15	mA	1-4
F27	Y1	28					
F28	Y2	32					
F29	Y3	30					
F30	Y4	86					
F31	Y5	85					
F32	Y6	87					
F33	Y7	88	0.0	-130	-15	mA	1-4
F34	-SEL B	20	0.0	-100	-15	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F34
Power supply current limit	0.51 A	NA	F1-F34
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F34
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F34
low, (VL)	0.0 V	± 100 mV	F1-F34
Output current measurement	NA	± 5.75 mA	F1-F4
	NA	± 5.0 mA	F5-F10
	NA	± 3.0 mA	F11-F13
	NA	± 4.25 mA	F14-F21
	NA	± 4.75 mA	F22,F23
	NA	± 3.0 mA	F24
	NA	± 4.25 mA	F25
	NA	± 5.75 mA	F26-F33
	NA	± 4.25 mA	F34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	ADDR1	2	2.7	0.0	20.0	uA	1-3
G2	ADDR2	3					
G3	ADDR3	4					
G4	ADDR4	6					
G5	A0	50					
G6	A1	77					
G7	A2	80					
G8	A3	82					
G9	A4	34					
G10	A5	33					
G11	A6	36					
G12	A7	38					
G13	B0	76					
G14	B1	78					
G15	B2	81					
G16	B3	79					
G17	B4	84					
G18	B5	83					
G19	B6	35					
G20	B7	37	2.7	0.0	20.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	-KADO	61	2.7	0.0	115.0	uA	1-3
G22	ADDR23	57	2.7	0.0	75.0	uA	1-3
G23	ADDR22	56	2.7	0.0	75.0	uA	1-3
G24	ADDR13	62	2.7	0.0	50.0	uA	1-3
G25	ADDR14	63					
G26	ADDR15	64	2.7	0.0	50.0	uA	1-3
G27	BES	5	2.7	0.0	130.0	uA	1-3
G28	-STAT EN0	14	2.4	0.0	25.0	uA	1-3
G29	-STAT EN1	9					
G30	READ	19	2.4	0.0	25.0	uA	1-3
G31	-KI	16	2.4	0.0	75.0	uA	1-3
G32	-RST	7	2.7	0.0	60.0	uA	1-3
G33	CLK	21	2.7	0.0	20.0	uA	1-3
G34	D0	44					
G35	D1	43					
G36	D2	42					
G37	D3	41					
G38	D4	48					
G39	D5	47					
G40	D6	46	2.7	0.0	20.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	D7	45	2.7	0.0	20.0	uA	1-3
G42	CTR CLK	29	2.7	0.0	100.0	uA	1-3
G43	CTR RST	26	2.7	0.0	20.0	uA	1-3
G44	-CTR RST	69					
G45	1A	23					
G46	1B	72					
G47	1C	22					
G48	1D	75					
G49	1E	74					
G50	1F	24					
G51	1G	73	2.7	0.0	20.0	uA	1-3
G52	3A	100	2.7	-1.08	-0.773	mA	1-3
G53	3B	99	2.7	0.0	20.0	uA	1-3
G54	FC0	53	2.7	0.0	50.0	uA	1-3
G55	FC1	52					
G56	FC2	51	2.7	0.0	50.0	uA	1-3
G57	BENB	59	2.4	0.0	75.0	uA	1-3
G58	-MUX OE	27	2.7	0.0	40.0	uA	1-3
G59	SEL B	70	2.7	0.0	100.0	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G59
Power supply current limit	0.51 A	NA	G1-G59
Forcing voltage, (Vih)	2.7 V	± 135 mV	G1-G27
	2.4 V	± 120 mV	G28-G31
	2.7 V	± 135 mV	G32-G56
	2.4 V	± 120 mV	G57
	2.7 V	± 135 mV	G58,G59
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G59
low, (VL)	0.0 V	± 100 mV	G1-G59
Input current measurement	NA	± 1.0 uA	G1-G20
	NA	± 5.75 uA	G21
	NA	± 3.75 uA	G22,G23
	NA	± 2.5 uA	G24-G26
	NA	± 6.5 uA	G27
	NA	± 1.25 uA	G28-G30
	NA	± 3.75 uA	G31
	NA	± 3.0 uA	G32
	NA	± 1.0 uA	G33-G41
	NA	± 5.0 uA	G42
	NA	± 1.0 uA	G43-G51
	NA	± 15.4 uA	G52
	NA	± 1.0 uA	G53
	NA	± 2.5 uA	G54-G56
	NA	± 3.75 uA	G57
	NA	± 2.5 uA	G58
	NA	± 5.0 uA	G59

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	ADDR1	2	0.4	-440.0	0.0	uA	1-3
H2	ADDR2	3					
H3	ADDR3	4					
H4	ADDR4	6					
H5	A0	50					
H6	A1	77					
H7	A2	80					
H8	A3	82					
H9	A4	34					
H10	A5	33					
H11	A6	36					
H12	A7	38					
H13	B0	76					
H14	B1	78					
H15	B2	81					
H16	B3	79					
H17	B4	84					
H18	B5	83					
H19	B6	35					
H20	B7	37	0.4	-440.0	0.0	uA	1-3

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Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	-KADO	61	0.4	-3.17	-1.07	mA	1-3
H22	ADDR23	57	0.4	-2.25	-1.0	mA	1-3
H23	ADDR22	56	0.4	-2.25	-1.0	mA	1-3
H24	ADDR13	62	0.5	-2.0	-1.0	mA	1-3
H25	ADDR14	63					
H26	ADDR15	64	0.5	-2.0	-1.0	mA	1-3
H27	BES	5	0.4	-2.76	-1.06	mA	1-3
H28	-STAT EN0	14	0.4	-250.0	-10.0	uA	1-3
H29	-STAT EN1	9					
H30	READ	19	0.4	-250.0	-10.0	uA	1-3
H31	-KI	16	0.4	-2.29	-1.03	mA	1-3
H32	-RST	7	0.4	-800.0	-60.0	uA	1-3
H33	CLK	21	0.4	-400.0	-30.0	uA	1-3
H34	D0	44	0.4	-400.0	-100.0	uA	1-3
H35	D1	43					
H36	D2	42					
H37	D3	41					
H38	D4	48					
H39	D5	47					
H40	D6	46	0.4	-400.0	-100.0	uA	1-3

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Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	D7	45	0.4	-400.0	-100.0	uA	1-3
H42	CTR CLK	29	0.4	-2.4	-0.35	mA	1-3
H43	CTR RST	26	0.4	-400.0	-30.0	uA	1-3
H44	-CTR RST	69	0.4	-400.0	-30.0	uA	1-3
H45	1A	23	0.4	-380.0	-30.0	uA	1-3
H46	1B	72					
H47	1C	22					
H48	1D	75					
H49	1E	74					
H50	1F	24					
H51	1G	73	0.4	-380.0	-30.0	uA	1-3
H52	3A	100	0.4	-2.36	-1.80	mA	1-3
H53	3B	99	0.4	-400.0	-30.0	uA	1-3
H54	FC0	53	0.5	-2.0	-1.0	mA	1-3
H55	FC1	52					
H56	FC2	51	0.5	-2.0	-1.0	mA	1-3
H57	BENB	59	0.4	-2.29	-1.03	mA	1-3
H58	-MUX OE	27	0.4	-880.0	0.0	uA	1-3
H59	SEL B	70	0.4	-2.16	0.0	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H59
Power supply current limit	0.51 A	NA	H1-H59
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H23
	0.5 V	± 25 mV	H24-H26
	0.4 V	± 20 mV	H27-H53
	0.5 V	± 25 mV	H54-H56
	0.4 V	± 20 mV	H57-H59
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H59
low, (VL)	0.0 V	± 100 mV	H1-H59
Input current measurement	NA	± 22.0 uA	H1-H20
	NA	± 105.0 uA	H21
	NA	± 62.5 uA	H22,H23
	NA	± 50.0 uA	H24-H26
	NA	± 85.0 uA	H27
	NA	± 12.0 uA	H28-H30
	NA	± 63.0 uA	H31
	NA	± 37.0 uA	H32
	NA	± 18.5 uA	H33
	NA	± 15.0 uA	H34-H41
	NA	± 102.0 uA	H42
	NA	± 18.5 uA	H43,H44
	NA	± 17.5 uA	H45-H51
	NA	± 28.0 uA	H52
	NA	± 18.5 uA	H53
	NA	± 50.0 uA	H54-H56

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 63.0 uA	H57
	NA	± 44.0 uA	H58
	NA	± 108.0 uA	H59

2. The following procedure is required to perform this test.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	TASK ID0	58	-1.0	2.4	4.5	V	1-4
J2	TASK ID1	8					
J3	TASK ID2	60					
J4	TASK ID3	11	-1.0	2.4	4.5	V	1-4
J5	-CS0	17	-2.0	2.4	4.5	V	1-4
J6	-CS1	67					
J7	-CS4	18					
J8	KI	68					
J9	DATA DIR	65					
J10	-STAT EN	66	-2.0	2.4	4.5	V	1-4
J11	-DATA EN	55	-1.0	2.5	4.5	V	1-4
J12	-CS2	13					
J13	-CS3	12	-1.0	2.5	4.5	V	1-4
J14	Q0	93	-0.4	2.5	4.5	V	1-4
J15	Q1	92					
J16	Q2	91					
J17	Q3	90					
J18	Q4	97					
J19	Q5	96					
J20	Q6	95	-0.4	2.5	4.5	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	Q7	94	-0.4	2.5	4.5	V	1-4
J22	1H	71					
J23	3C	98	-0.4	2.5	4.5	V	1-4
J24	-IACK	54	-1.0	2.5	4.5	V	1-4
J25	-CTR Q7	49	-0.4	2.5	4.5	V	1-4
J26	Y0	39	-1.0	2.4	4.5	V	1-4
J27	Y1	28					
J28	Y2	32					
J29	Y3	30					
J30	Y4	86					
J31	Y5	85					
J32	Y6	87					
J33	Y7	88	-1.0	2.4	4.5	V	1-4
J34	-SEL B	20	-0.4	2.5	4.5	V	1-4
J35	PU	89	-2.0	2.4	2.6	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J35
Power supply current limit	0.51 A	NA	J1-J35
Forcing current, (Ioh)	-1.0 mA	± 100 uA	J1-J4
	-2.0 mA	± 200 uA	J5-J10
	-1.0 mA	± 100 uA	J11-J13
	-0.4 mA	± 40 uA	J14-J23
	-1.0 mA	± 100 uA	J24
	[illegible]	[illegible]	J25
	-1.0 mA	± 100 uA	J26-J33
	-0.4 mA	± 40 uA	J34
	-2.0 mA	± 200 uA	J35
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J35
low, (VL)	0.0 V	± 100 mV	J1-J35
Output voltage measurement	NA	± 105 mV	J1-J10
	NA	± 100 mV	J11-J25
	NA	± 105 mV	J26-J33
	NA	± 100 mV	J34
	NA	± 10 mV	J35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

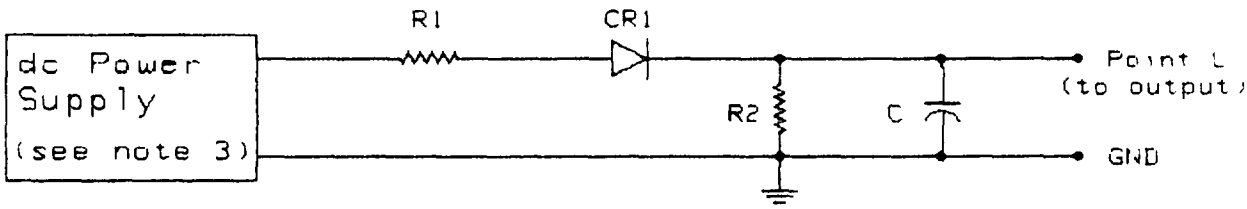
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 8. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are in the table below.

5. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
8,11,28,30,32,39, 58,60,85,86,87, 88	2400	12	0.4
20,49,71,90,91, 92,93,94,95,96, 97,98	6250	4	0.4
17,18,65,66,67, 68	1200	8	0.5
12,13,54,55	2500	20	0.5
89	1200	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	TASK ID0	58	12.0	0.0	0.4	V	1-4
K2	TASK ID1	8					
K3	TASK ID2	60					
K4	TASK ID3	11	12.0	0.0	0.4	V	1-4
K5	-CS0	17	8.0	0.0	0.5	V	1-4
K6	-CS1	67					
K7	-CS4	18					
K8	KI	68					
K9	DATA DIR	65					
K10	-STAT EN	66	8.0	0.0	0.5	V	1-4
K11	-DATA EN	55	20.0	0.2	0.5	V	1-4
K12	-CS2	13	20.0	0.0	0.5	V	1-4
K13	-CS3	12	20.0	0.0	0.5	V	1-4
K14	Q0	93	4.0	0.0	0.4	V	1-4
K15	Q1	92					
K16	Q2	91					
K17	Q3	90					
K18	Q4	97					
K19	Q5	96					
K20	Q6	95	4.0	0.0	0.4	V	1-4

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Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	Q7	94	4.0	0.0	0.4	V	1-4
K22	1H	71					
K23	3C	98	4.0	0.0	0.4	V	1-4
K24	-1ACK	54	20.0	0.2	0.5	V	1-4
K25	-CTR Q7	49	4.0	0.0	0.4	V	1-4
K26	Y0	39	12.0	0.0	0.4	V	1-4
K27	Y1	28					
K28	Y2	32					
K29	Y3	30					
K30	Y4	86					
K31	Y5	85					
K32	Y6	87					
K33	Y7	88	12.0	0.0	0.4	V	1-4
K34	-SEL B	20	4.0	0.0	0.4	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K34
Power supply current limit	0.51 A	NA	K1-K34
Forcing current, (Iol)	12.0 mA	± 1.2 mA	K1-K4
	8.0 mA	± 0.8 mA	K5-K10
	20.0 mA	± 2.0 mA	K11-K13
	4.0 mA	± 0.4 mA	K14-K23
	20.0 mA	± 2.0 mA	K24
	4.0 mA	± 0.4 mA	K25
	12.0 mA	± 1.2 mA	K26-K33
	4.0 mA	± 0.4 mA	K34
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K34
low, (VL)	0.0 V	± 100 mV	K1-K34
Output voltage measurement	NA	± 20 mV	K1-K4
	NA	± 25 mV	K5-K10
	NA	± 15 mV	K11
	NA	± 25 mV	K12,K13
	NA	± 20 mV	K14-K23
	NA	± 15 mV	K24
	NA	± 20 mV	K25-K34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	Y0	39	2.7	-20	20	uA	1-3
L2	Y1	28					
L3	Y2	32					
L4	Y3	30					
L5	Y4	86					
L6	Y5	85					
L7	Y6	87					
L8	Y7	88	2.7	-20	20	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L8
Power supply current limit	0.51 A	NA	L1-L8
Forcing voltage, (Voh)	2.7 V	± 135 mV	L1-L8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 L1-L8 L1-L8
Output current measurement	NA	± 2 uA	L1-L8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	Y0	39	0.4	-20	20	uA	1-3
M2	Y1	28					
M3	Y2	32					
M4	Y3	30					
M5	Y4	86					
M6	Y5	85					
M7	Y6	87					
M8	Y7	88	0.4	-20	20	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M8
Power supply current limit	0.51 A	NA	M1-M8
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 M1-M8 M1-M8
Output current measurement	NA	± 2 uA	M1-M8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions of table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	2	34	340	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.51 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 15.3 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.51 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 23.1 mV ± 34.1 mV ± 5.5 mV	 P1 P2 P1,P2

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4.8.15 - Continued.

4.8.15.1 Following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 150 pF (see figure 8). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

3 Abbreviations used are defined in 6.1.

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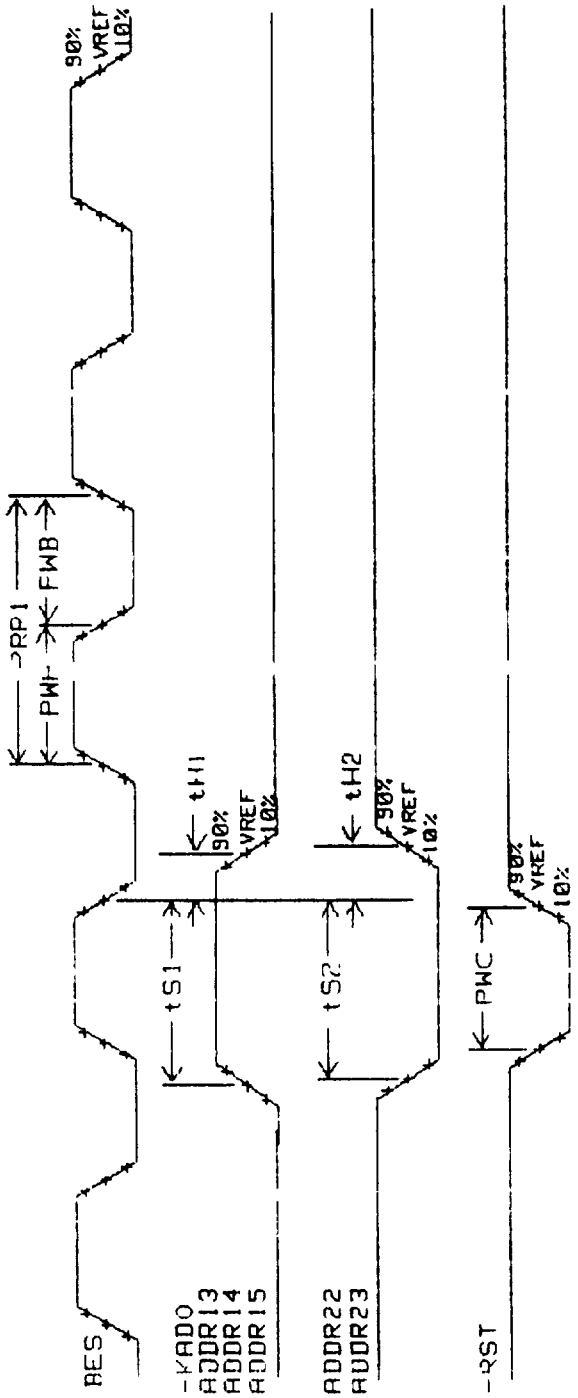


FIGURE 9. Input timing parameters for table XV.

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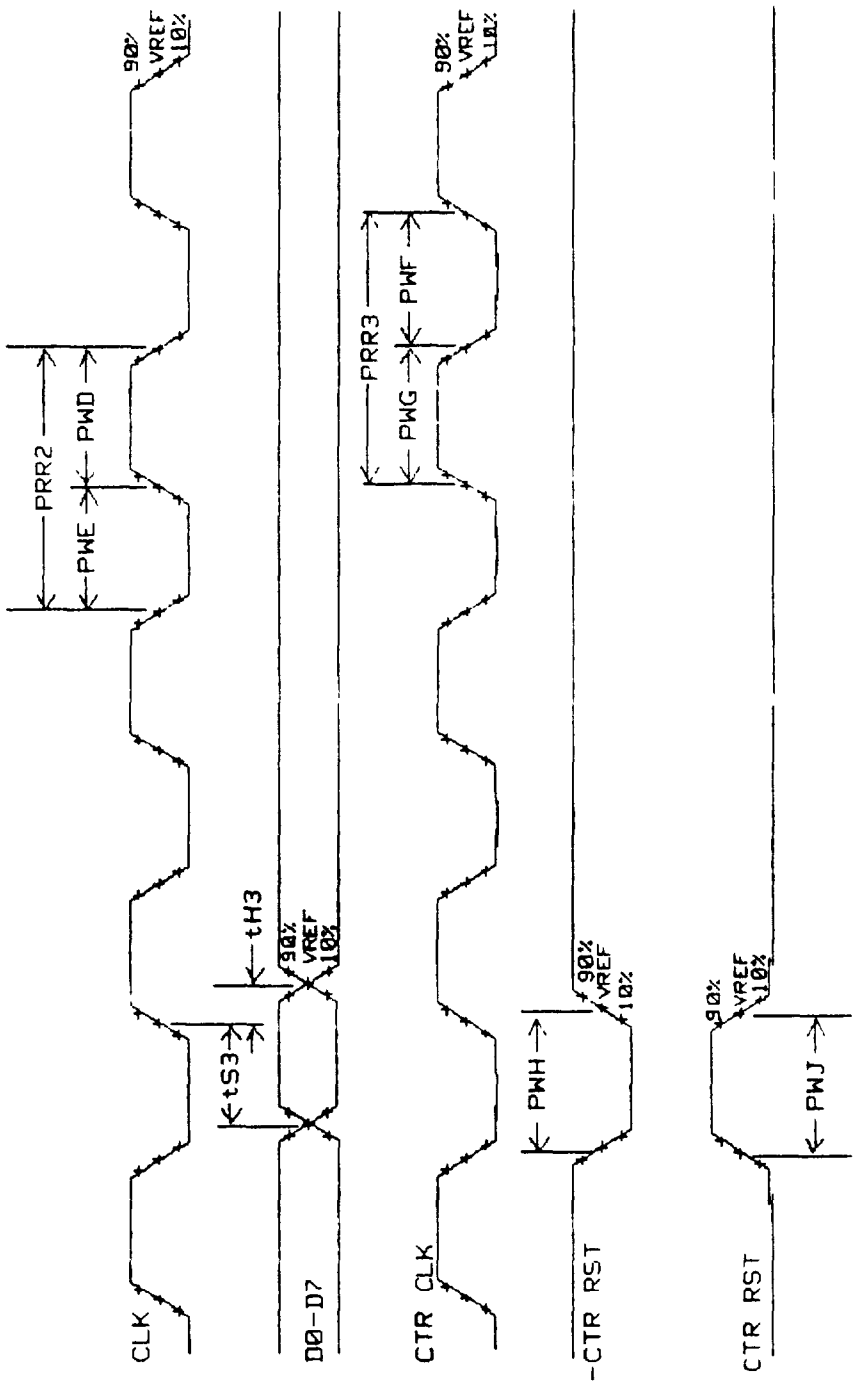


FIGURE 9. Input timing parameters for table XV - Continued.

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NOTES:

1. Input parameters for figure 9.

PRR1 = 200 nS minimum.
PRR2 = 100 nS minimum.
PRR3 = 100 nS minimum.
PWA = 100 nS minimum.
PWB = 100 nS minimum.
PWC = 50 nS minimum.
PWD = 50 nS minimum.
PWE = 50 nS minimum.
PWF = 50 nS minimum.
PWG = 50 nS minimum.
PWH = 75 nS minimum.
PWJ = 53 nS minimum.
TS1 = 61 nS minimum.
TS2 = 61 nS minimum.
TS3 = 20 nS minimum.
TH1 = 0 nS minimum.
TH2 = 0 nS minimum.
TH3 = 5 nS minimum.
VREF = 1.3 V $\pm$ 0.1 V.

2. Pattern voltages shall be as specified in the test tolerance table.

3. Rise times (as measured between the 10 percent and 90 percent points) shall be 15 nS maximum.

4. Fall times (as measured between the 90 percent and 10 percent points) shall be 6 nS maximum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1 Q2
Power supply current limit	0.51 A	NA	Q1,Q2
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1,Q2 Q1,Q2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 23.1 mV ± 34.1 mV ± 5.5 mV	 Q1 Q2 Q1,Q2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

3. The load circuit for this test is specified on figure 8.

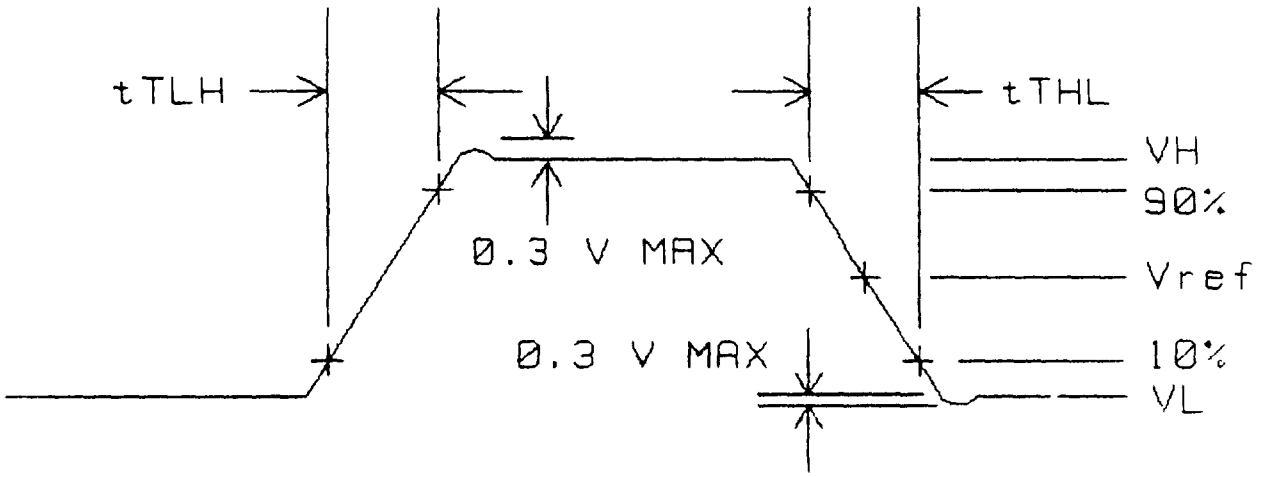
4. All of the input waveforms shall meet the requirements shown on figure 10.

5. Input/Output (I/O) waveforms are as referenced on figure 11.

6. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
2,3,4,6,7,21,22,23,24,26,27, 29,33,34,35,36,37,38,41,42, 43,44,45,46,47,48,50,69,70, 72,73,74,75,76,77,78,79,80, 81,82,83,84,99,100	15 ± 1.5 nS	6 ± 1.5 nS
5,9,14,16,19,51,52,53,56,57, 59,61,62,63,64	2.5 ± 1.5 nS	2.5 ± 1.5 nS

FIGURE 10. Sample input waveform for table XVI.

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TABLE XVII. Dynamic parameters.

Ref no	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
47	1-8	CLK to Q0-Q7 tplh	-	-	-	54	44	54	nS	11h
48	9-16	CLK to Q0-Q7 tphl	-	-	-	53	43	53	nS	11h
10	17-24	B0-B7 to Y0-Y7 tplh	-	-	-	42	30	42	nS	11a
9	25-32	B0-B7 to Y0-Y7 tphl	-	-	-	39	27	39	nS	11a
8	33-40	A0-A7 to Y0-Y7 tplh	-	-	-	42	30	42	nS	11a
7	41-48	A0-A7 to Y0-Y7 tphl	-	-	-	39	27	39	nS	11a
12	49-56	SEL B(LH) to Y0-Y7tplh	-	-	-	46	33	46	nS	11a
11	57-64	SEL B(HL) to Y0-Y7tphl	-	-	-	43	30	43	nS	11a
13	65	SEL B to -SEL B tplh	-	-	-	32	27	32	nS	11a
14	66	SEL B to -SEL B tphl	-	-	-	35	28	35	nS	11a
49	67-74	D0-D7 Setup Time tsu	20	20	20	-	-	-	nS	11h
50	75-82	D0-D7 Holdt me th	5	5	5	-	-	-	nS	11h
45	83-90	CLK Pulse Width LOPWL	50	50	50	-	-	-	nS	11h
44	91-98	CLK Max Freq Fmax	10	10	10	-	-	-	MHz	11h
46	99-106	CLK Pulse Width HIPWH	50	50	50	-	-	-	nS	11h
61	107-114	-MUX OE to Y0-Y7 tpzl	-	-	-	57	39	57	nS	11g

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TABLE XVII. Dynamic parameters - Continued.

Ref no.	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
60	115-122	-MUX OE to Y0-Y7 tpzh	-	-	-	60	42	60	nS	11h
51	123-125	1A-1C to 1H tplh	-	-	-	37	27	37	nS	11h
52	126-128	1A-1C to 1H tphl	-	-	-	41	31	41	nS	11h
51	129-132	1D-1G to 1H tplh	-	-	-	62	42	62	nS	11h
52	133-136	1D-1G to 1H tphl	-	-	-	71	51	71	nS	11h
16	137-138	3A,3B to 3C tplh	-	-	-	47	32	47	nS	11c
15	139-140	3A,3B to 3C tphl	-	-	-	46	31	46	nS	11c
18	141-144	BENB,FC0,1,2 to -IACK tplh	-	-	-	15	13	15	nS	11c
17	145-148	BENB,FC0,1,2 to -IACK tphl	-	-	-	13	11 5	13	nS	11c
41	149	CTR CLk to -CTR Q7tplh	-	-	-	200	157	200	nS	11g
40	150	CTP CLK to -CTR Q7tphl	-	-	-	201	158	201	nS	11g
43	151	CTP RST to -CTR Q7tplh	-	-	-	118	86	118	nS	11g
42	152	-CTR RST to -CTR Q7 tplh	-	-	-	138	101	138	nS	11g
35	153	CTR CLK PWH	50	50	50	-	-	-	nS	11g

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TABLE XVII. Dynamic parameters - Continued.

Ref no	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
34	154	CTR CLK PWL	50	50	50	-	-	-	nS	11g
30, 31	155-158	ADDR1-4 to TASK ID0-3 tplh	-	-	-	42	30	42	nS	11f
30, 31	159-162	ADDR1-4 to TASK ID0-3 tphl	-	-	-	39	27	39	nS	11f
62	163-166	-KADO to TASK ID0-3 tplh	-	-	-	46	33	46	nS	11f
32	167-170	-KADO to TASK ID0-3 tphl	-	-	-	43	30	43	nS	11f
2	171-172	ADDR13,15 to -CS2 tphl	-	-	-	20	20	20	nS	11b
1	173-174	ADDR13,15 to -CS2 tplh	-	-	-	18	18	18	nS	11b
1	175-176	ADDR14 to -CS2, -CS3 tplh	-	-	-	23	23	23	nS	11b
2	177-178	ADDR14 to -CS2, -CS3 tphl	-	-	-	20	20	20	nS	11b
1	179	ADDR13 to -CS3 tplh	-	-	-	23	23	23	nS	11b
2	180	ADDR13 to -CS3 tphl	-	-	-	20	20	20	nS	11b
2	181	ADDR15 to -CS3 tphl	-	-	-	20	20	20	nS	11b
1	182	ADDR15 to -CS3 tplh	-	-	-	18	18	18	nS	11b

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TABLE XVII. Dynamic parameters - Continued

Ref no	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
3	183-184	-KADO to -CS2, -CS3 tplh	-	-	-	22	22	22	nS	11b
4	185-186	-KADO to -CS2, -CS3 tphl	-	-	-	20	20	20	nS	11b
2	187-190	ADDR22,23 to -CS2, -CS3 tphl	-	-	-	20	20	20	nS	11b
1	191-194	ADDR22,23 to -CS2 -CS3 tplh	-	-	-	18	18	18	nS	11b
3	195-196	-KADO to -CS0, -CS1 tplh	-	-	-	68	68	68	nS	11b
4	197-198	-KADO to -CS0, -CS1 tphl	-	-	-	72	72	72	nS	11b
5	199	-KADO to -CS4 tphl	-	-	-	56	56	56	nS	11b
6	200	-KADO to -CS4 tplh	-	-	-	52	52	52	nS	11b
1	201-206	ADDR22,23 to -CS0, -CS1,-CS4 tplh	-	-	-	64	64	64	nS	11b
2	207-212	ADDR22,23 to -CS0, -CS1,-CS4 tphl	-	-	-	72	72	72	nS	11b

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TABLE XVII. Dynamic parameters - Continued.

Ref rc	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
1	213-216	HDDR13,14 to -CS0, -CS4 tplh	-	-	-	64	64	64	nS	11b
1	217	ADDR13 to -CS1 tplh	-	-	-	69	69	69	nS	11b
2	218-221	HDDR13,14 to -CS0, -CS4 tphl	-	-	-	72	72	72	nS	11b
1	222-223	ADDR14,15 to -CS1 tplh	-	-	-	64	64	64	nS	11b
1	224	ADDR15 to -CS0 tplh	-	-	-	64	64	64	nS	11b
1	225	ADDR15 to -CS4 tplh	-	-	-	69	69	69	nS	11b
2	226	ADDR13 to -CS1 tphl	-	-	-	72	72	72	nS	11b
2	227-228	ADDR14,15 to -CS1 tphl	-	-	-	72	72	72	nS	11b
	229	BES FMAX, BES Pulse Width Hi + BES Pulse Width Lo	5	5	5	-	-	-	MHz	11e, 11k
22	230	RST to -CS1 tphl	-	-	-	96	84	96	nS	11d
22	231	-RST to -CS0 tplh	-	-	-	92	80	92	nS	11d
26	232	READ to DATA DIR tplh	-	-	-	52	52	52	nS	11e
27	233	READ to DATA DIR tphl	-	-	-	56	56	56	nS	11e
28	234	BENB to DATA DIR tplh	-	-	-	52	52	52	nS	11e
29	235	BENB to DATA DIR tphl	-	-	-	56	56	56	nS	11e

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TABLE XVII. Dynamic parameters - Continued.

Ref no.	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
24	236-237	ADDR22,23 to DATA DIR tplh	-	-	-	52	52	52	nS	11e
25	238-239	ADDR22,23 to DATA DIR tphl	-	-	-	56	56	56	nS	11e
16	240-241	-STAT EN0,-STAT EN1 to -STAT EN tplh	-	-	-	52	52	52	nS	11c
15	242-243	-STAT EN0,-STAT EN1 to -STAT EN tphl	-	-	-	56	56	56	nS	11c
59	244	-KI to KI tplh	-	-	-	52	52	52	nS	11i
58	245	-KI to KI tphl	-	-	-	56	56	56	nS	11i
54, 56	246-247	BES,-KI to -DATA EN tplh	-	-	-	15	13	15	nS	11i
55, 57	248-249	BES,-KI to -DATA EN tphl	-	-	-	13	11.5	13	nS	11i
65	250	-KADO to BES SETUP tsu	61	56	61	-	-	-	nS	11d
63	251-252	ADDR22,23 to BES SETUP tsu	61	56	61	-	-	-	nS	11d

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TABLE XVII. Dynamic parameters - Continued.

Ref no.	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
19	253-255	ADDR13,14,15 to BES SETUP tsu	61	56	61	-	-	-	nS	11d
66	256	-KADO to BES HOLD th	0	0	0	-	-	-	nS	11d
64	257-258	ADDR22,23 to BES HOLD th	0	0	0	-	-	-	nS	11d
20	259-261	ADDR13,14,15 to BES HOLD th	0	0	0	-	-	-	nS	11d
21	262	BES PULSE WIDth HI PWH	100	100	100	-	-	-	nS	11d
53	263	BES PULSE WIDTH LO PWL	100	100	100	-	-	-	nS	11i
33	264	CTR CLK MAX FREQ FMAX	10	10	10	-	-	-	MHz	11g
37	265	CTR RST PULSE WIDTH HI PWH	100	100	100	-	-	-	nS	11g
36	266	-CTR RST PULSE WIDTH LO PWL	100	100	100	-	-	-	nS	11g
2	267	ADDR15 to -CS0 tphl	-	-	-	69	69	69	nS	11b
1	268	ADDR15 to -CS4 tplh	-	-	-	69	69	69	nS	11b

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TABLE XVII. Dynamic parameters - Continued.

Ref no.	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
67	269	-CTR RST to CTR CLK setup time tsu	90	68	90	-	-	-	nS	11g
68	270	CTR RST to CTR CLK setup time tsu	70	50	70	-	-	-	nS	11g

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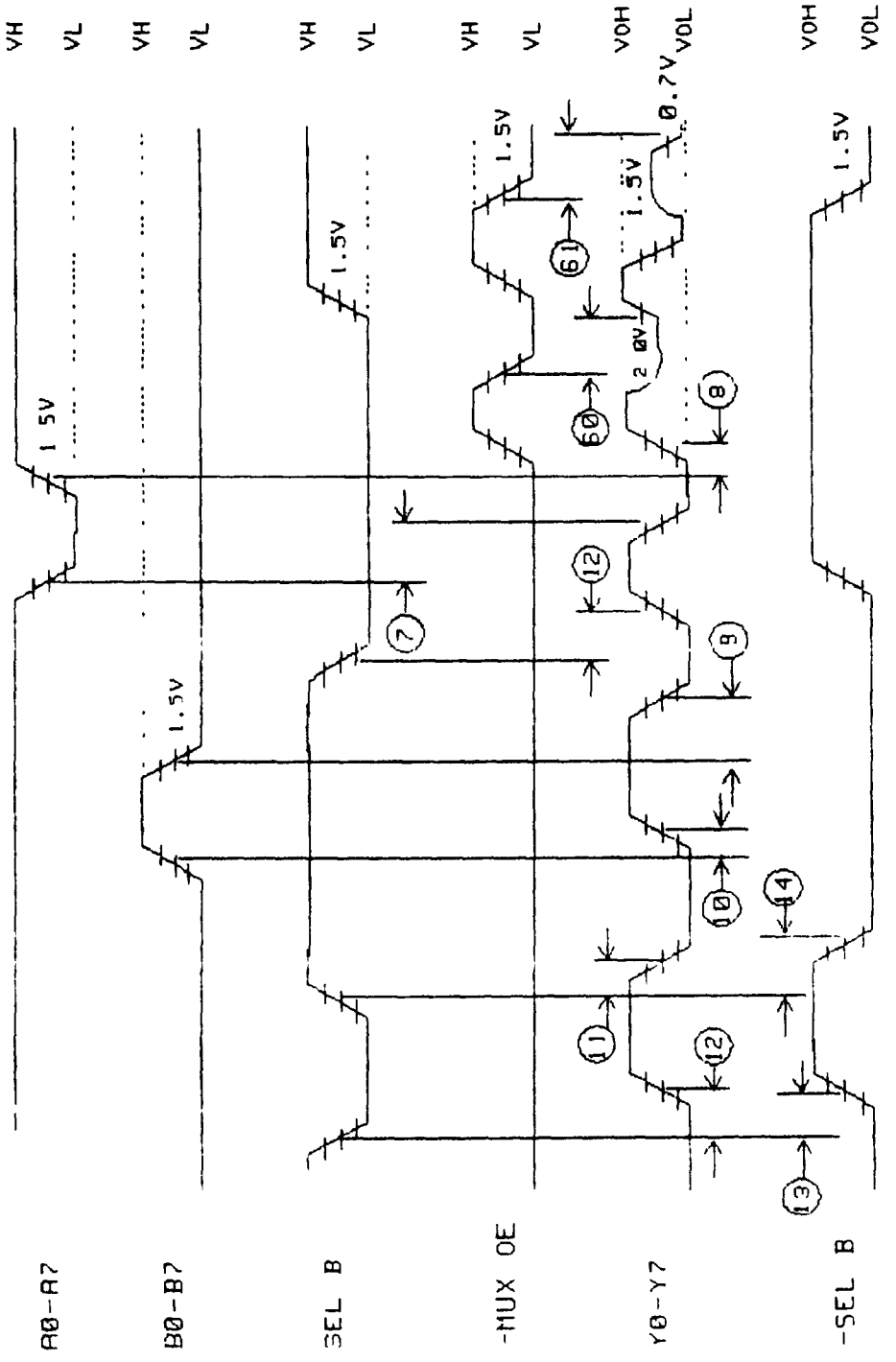


FIGURE 11a. Dynamic waveforms for table XVII.

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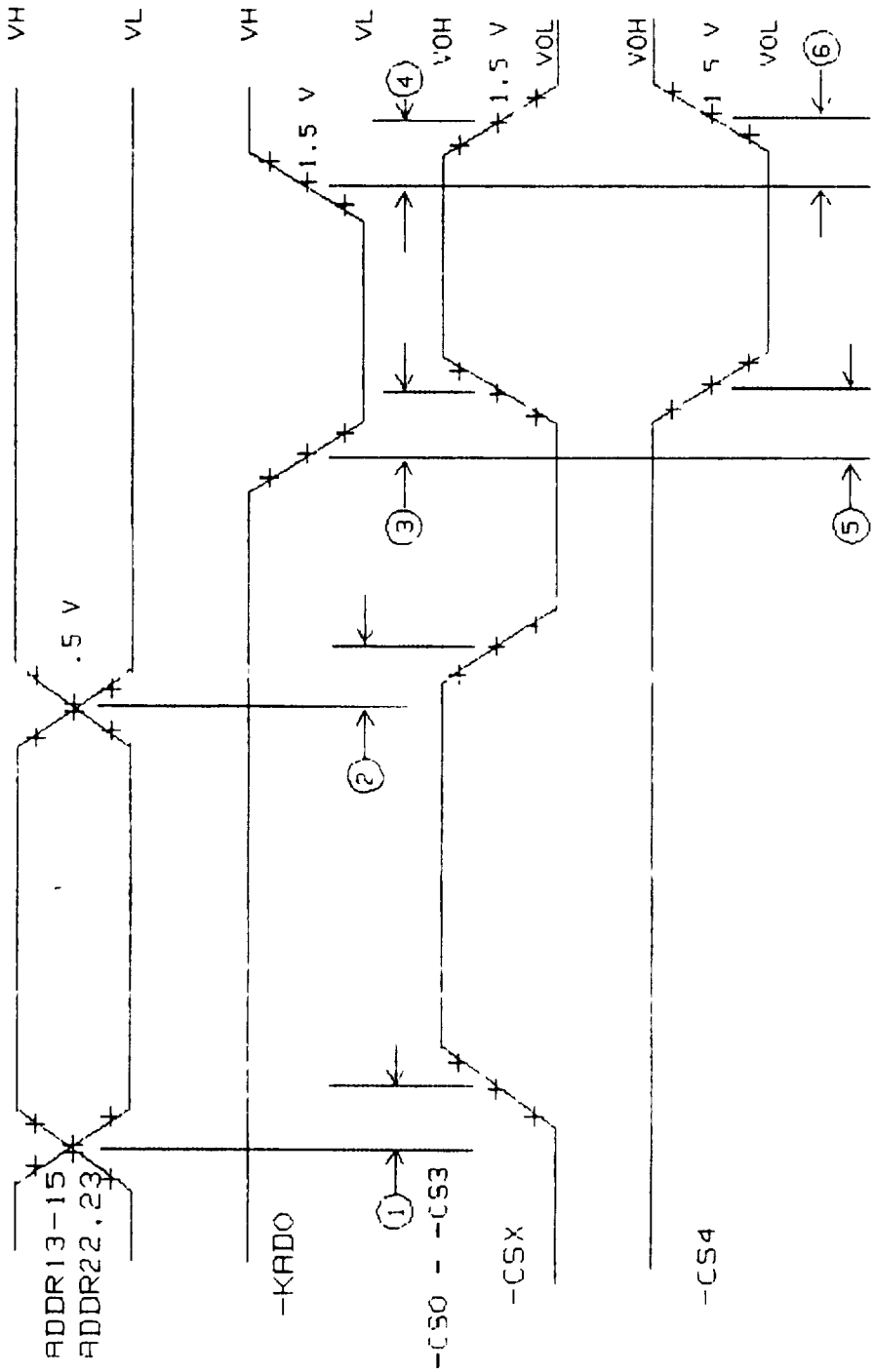


FIGURE 11b. Dynamic waveforms for table XVII.

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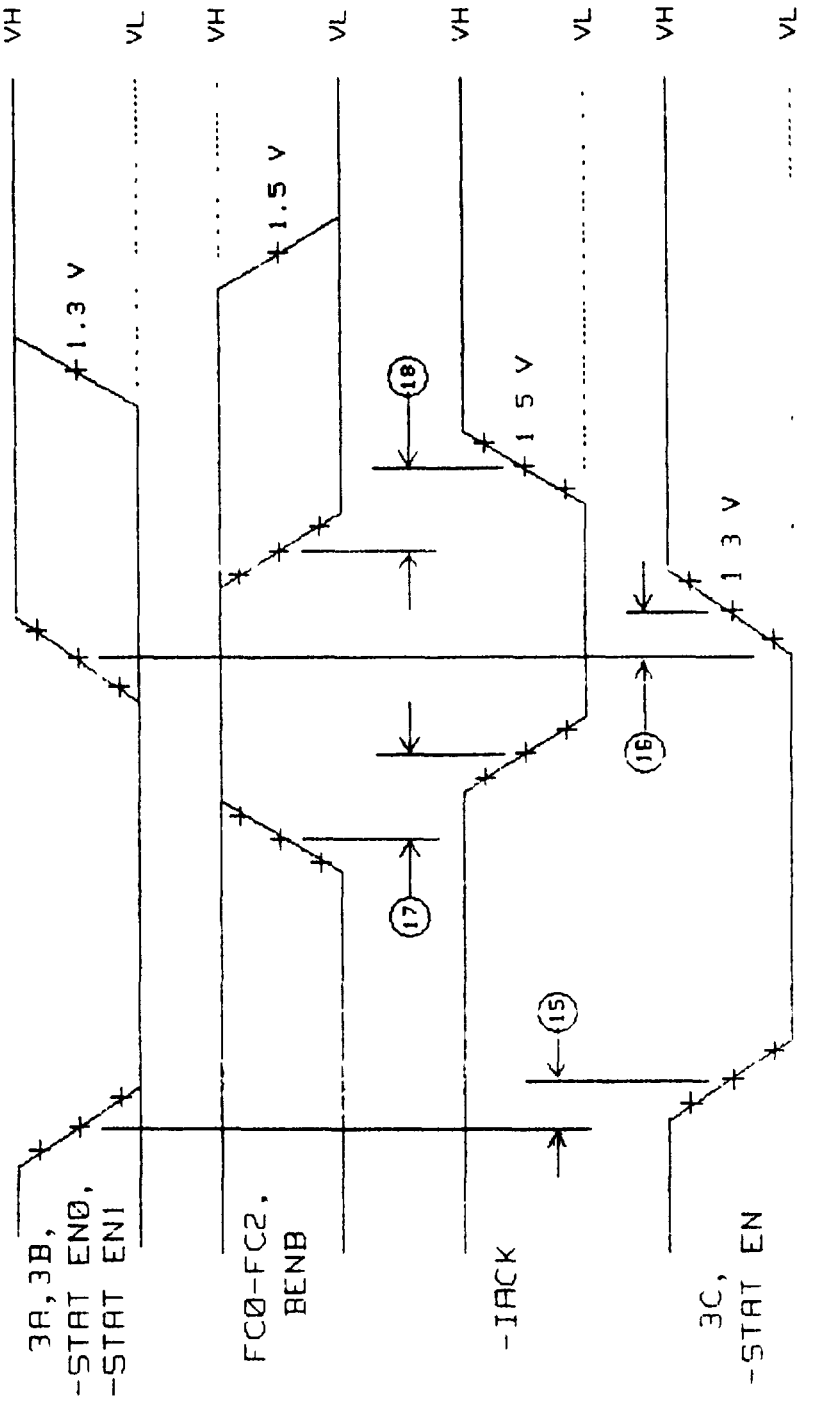


FIGURE 11c. Dynamic waveforms for table XVII.

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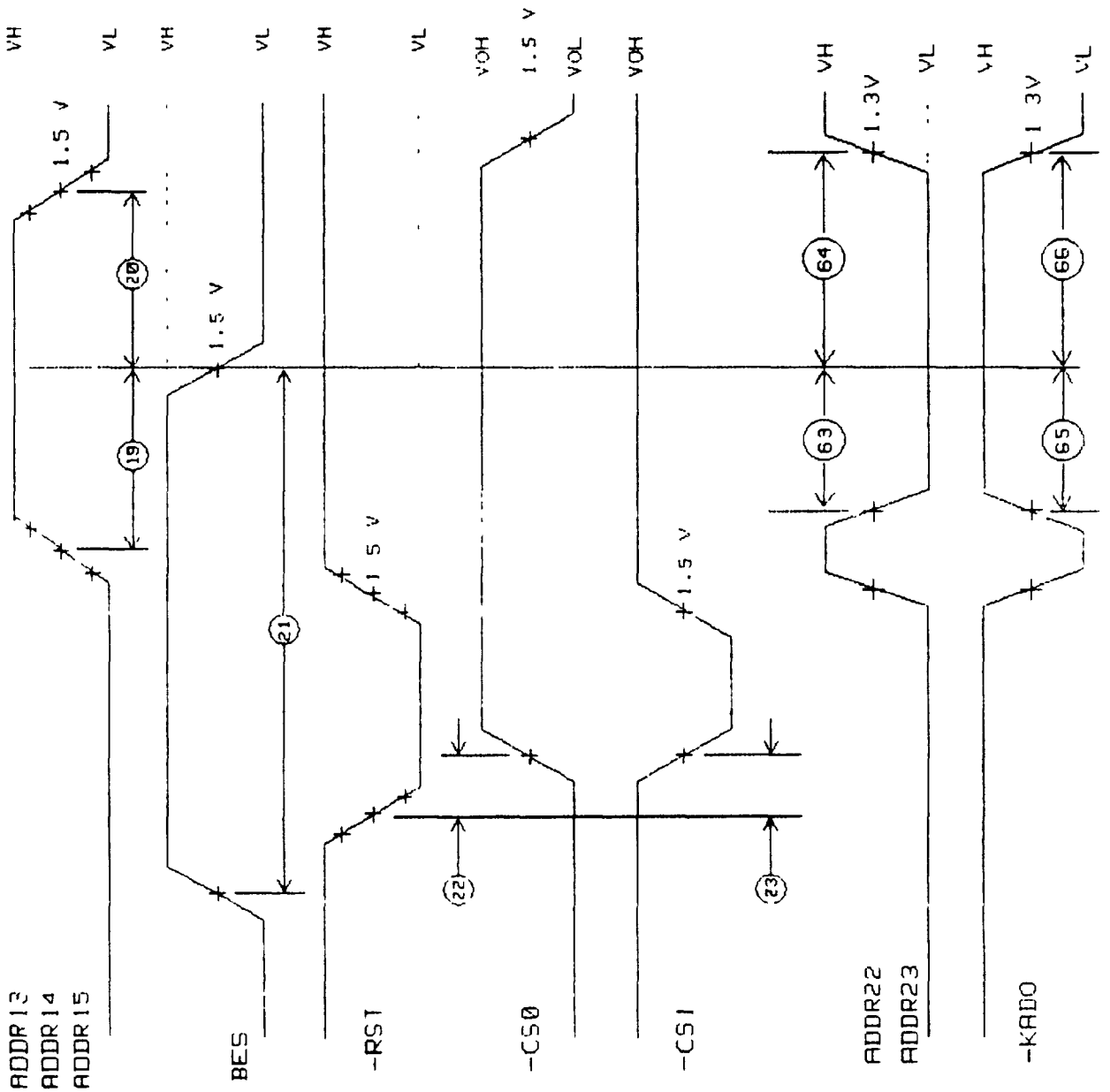


FIGURE 11d. Dynamic waveforms for table XVII.

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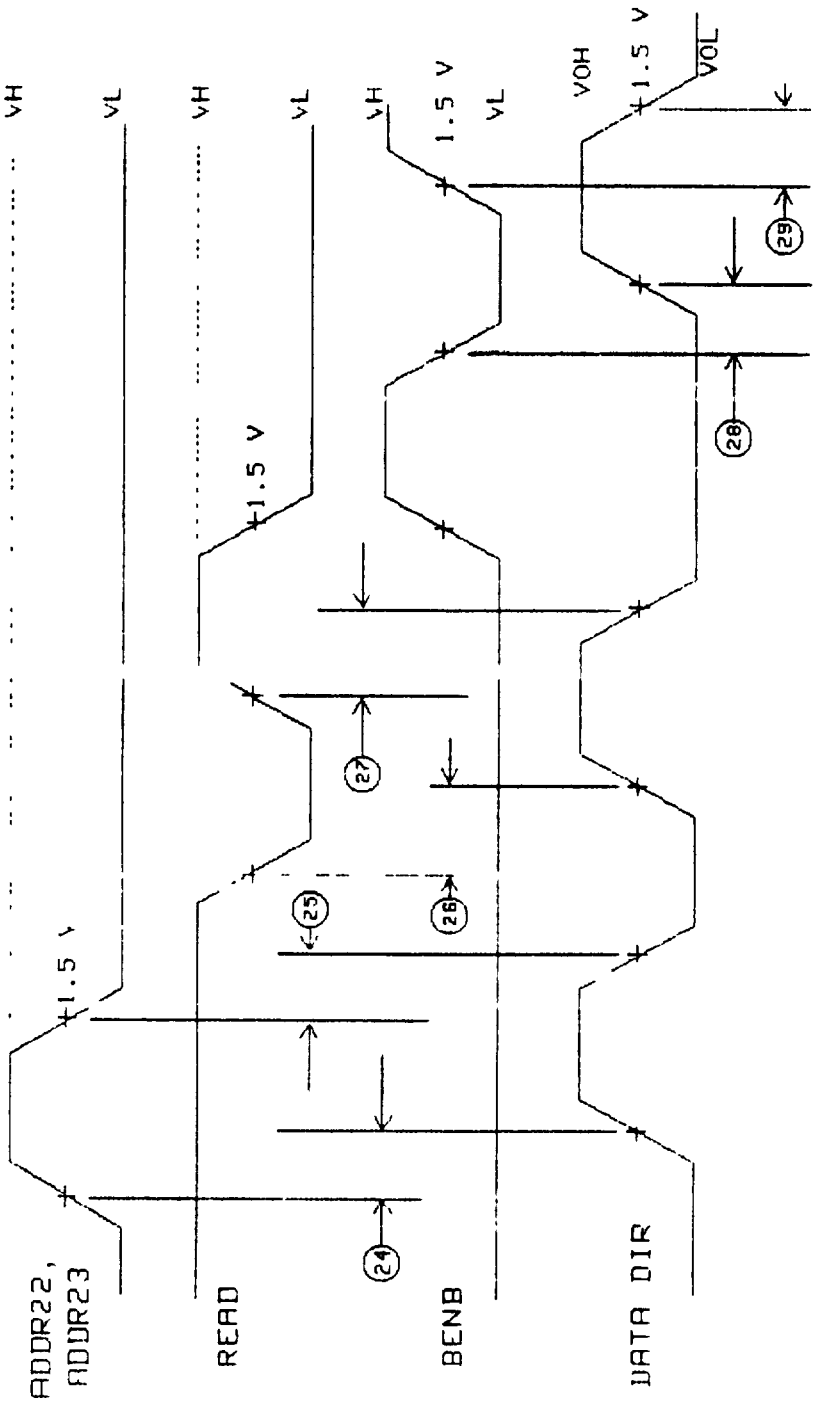


FIGURE 11e. Dynamic waveforms for table XVII.

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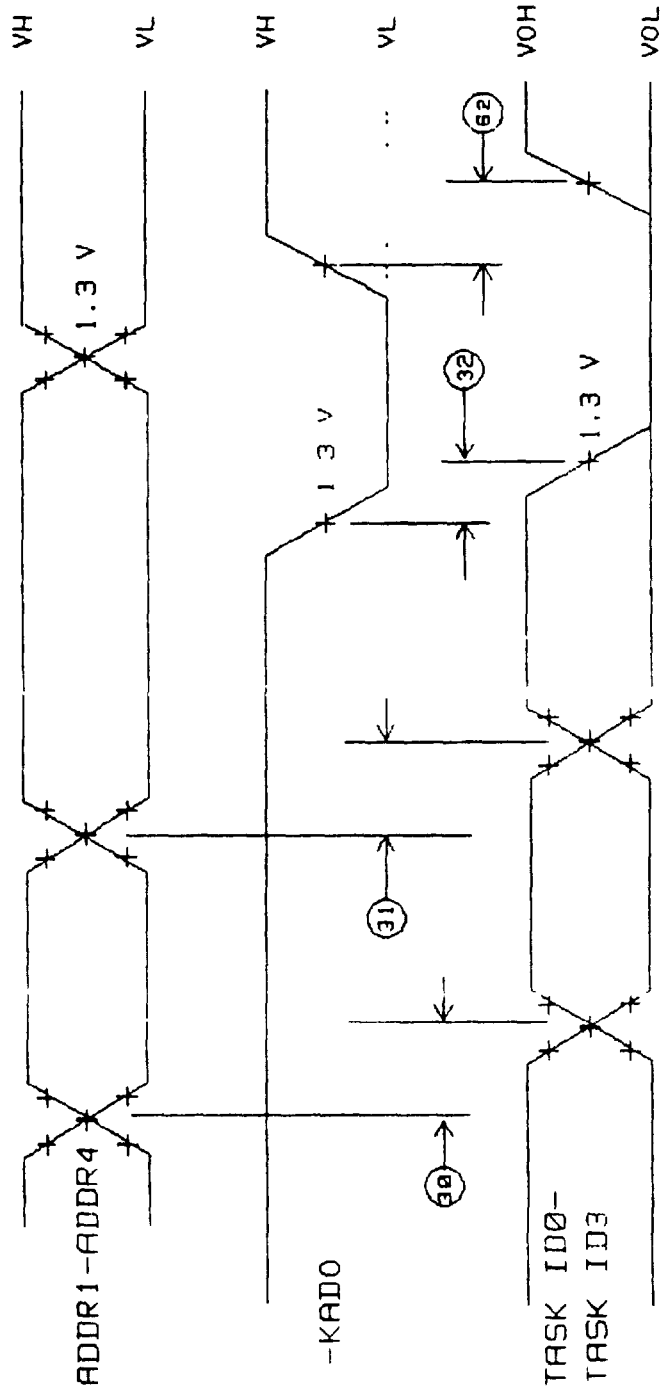


FIGURE 11f. Dynamic waveforms for table XVII

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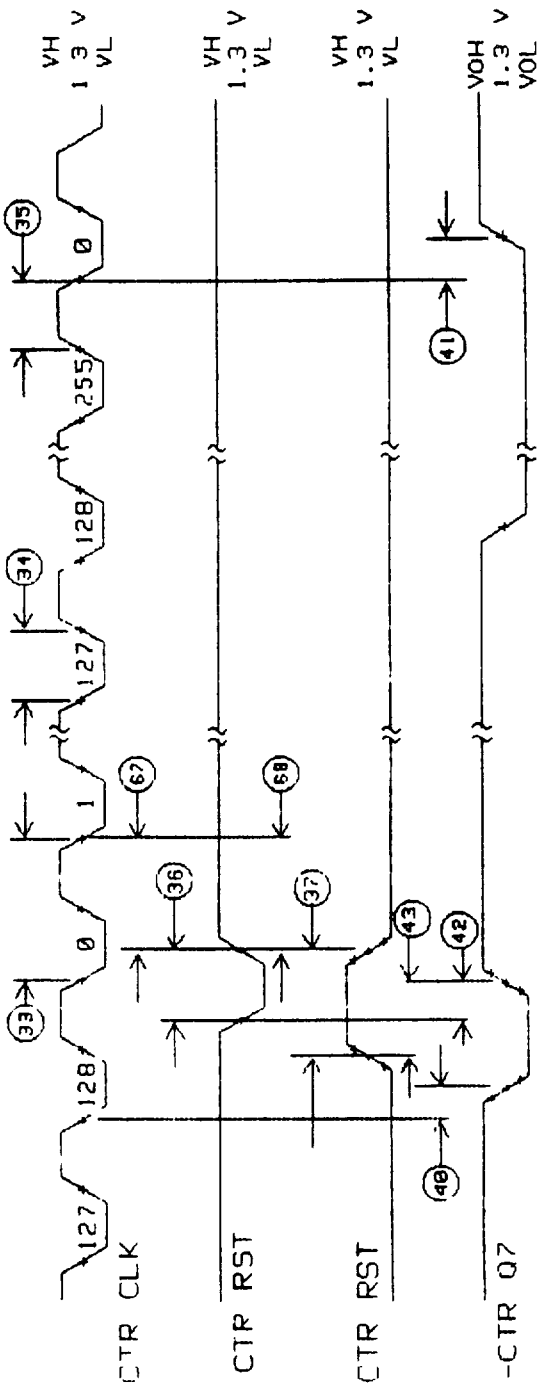


FIGURE 11g. Dynamic waveforms for table XVII.

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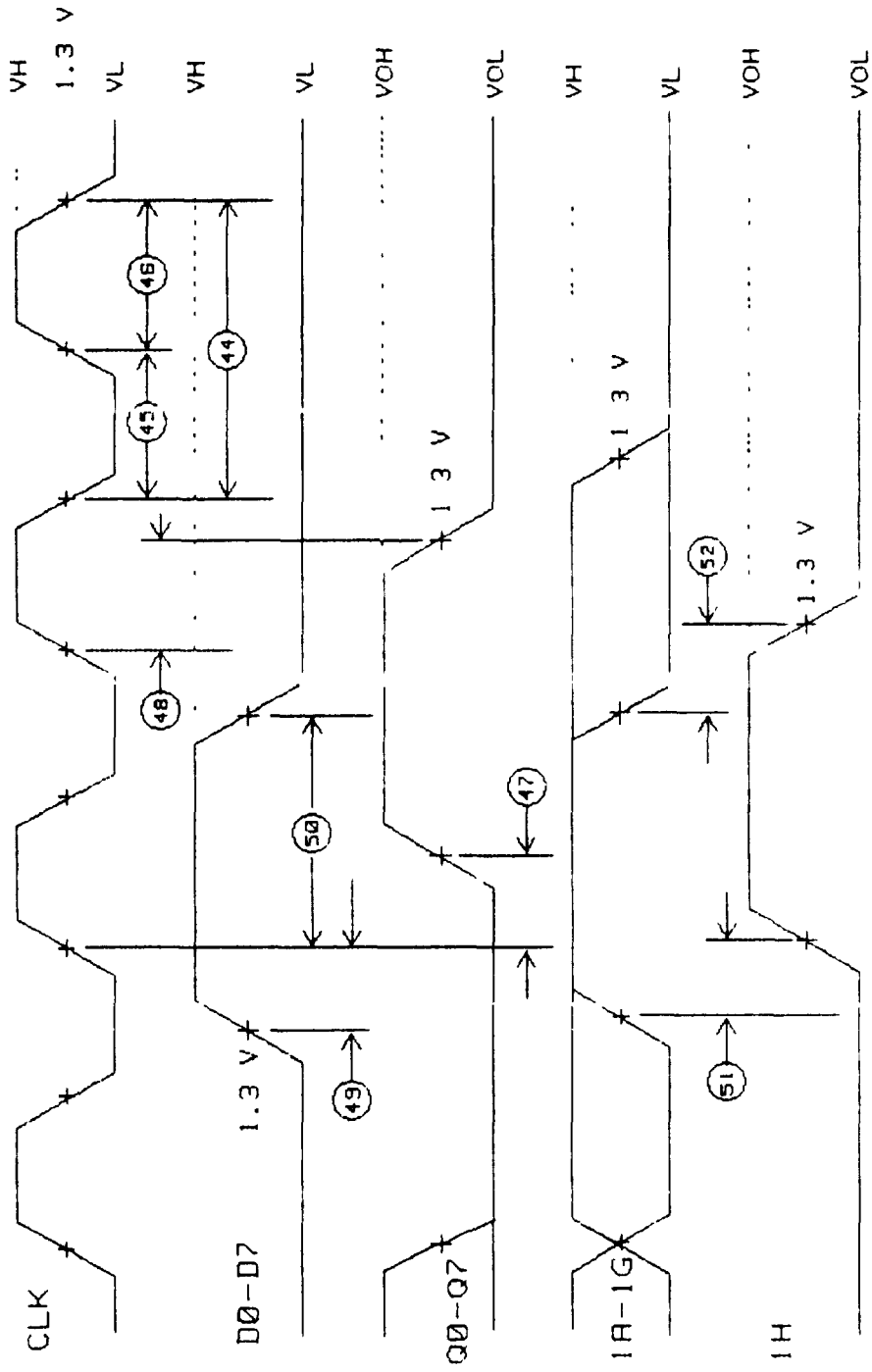


FIGURE 11h. Dynamic waveforms for table XVII.

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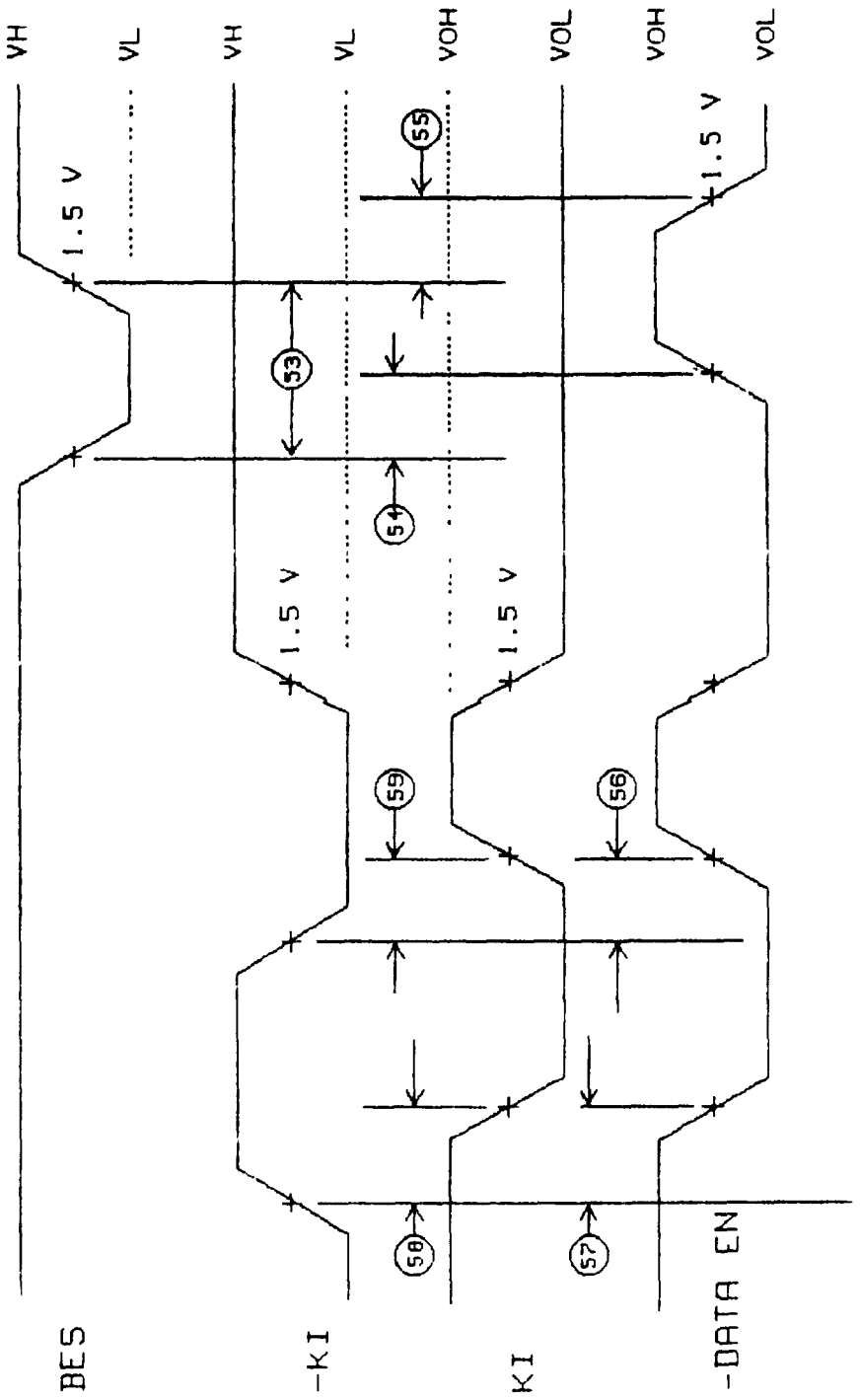


FIGURE 111. Dynamic waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.9	7.3	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 170 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

4. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1	0	400	mV	1-3

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	0.51 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX. The module inputs shall not violate the minimum timing parameters referenced on figure 9.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 12 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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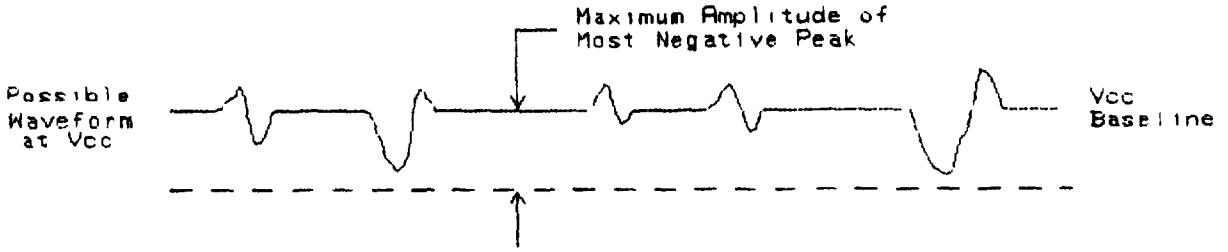


FIGURE 12. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
T1	TASK ID0	58	0	400	mV	1-5	13,14
T2	TASK ID1	8					
T3	TASK ID2	60					
T4	TASK ID3	11					
T5	-CS0	17					
T6	-CS1	67					
T7	-CS4	18					
T8	KI	68					
T9	DATA DIR	65					
T10	-STAT EN	66					
T11	-DATA EN	55					
T12	-CS2	13					
T13	-CS3	12					
T14	Q0	93					
T15	Q1	92					
T16	Q2	91					
T17	Q3	90					
T18	Q4	97					
T19	Q5	96					
T20	Q6	95	0	400	mV	1-5	13,14

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
T21	Q7	94	0	400	mV	1-5	13,14
T22	1H	71					
T23	3C	98					
T24	-IACK	54					
T25	-CTR Q7	49					
T26	Y0	39					
T27	Y1	28					
T28	Y2	32					
T29	Y3	30					
T30	Y4	86					
T31	Y5	85					
T32	Y6	87					
T33	Y7	88					
T34	-SEL B	20					
T35	PU	89	0	400	mV	1-5	13,14

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NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T35
Power supply current limit	0.51 A	NA	T1-T35
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T35 T1-T35
Output crosstalk measurement	NA	± 20 mV	T1-T35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. The test circuit shall be as indicated on figure 14. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

5. Abbreviations used are defined in 6.1.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
U1	1H	71	0	400	mV	1-5	13,14
U2	3C	98					
U3	-SEL B	20					
U4	-IACK	54					
U5	-CTR Q7	49	0	400	mV	1-5	13,14
U6	-DATA EN	55	0	500	mV	1-5	13,14
U7	KI	68					
U8	-CS0	17					
U9	-CS1	67					
U10	-CS2	13					
U11	-CS3	12					
U12	-CS4	18					
U13	DATA DIR	65					
U14	-STAT EN	66					
U15	Q0	93					
U16	Q1	92					
U17	Q2	91					
U18	Q3	90					
U19	Q4	97					
U20	Q5	96	0	500	mV	1-5	13,14

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
U21	Q6	95	0	500	mV	1-5	13,14
U22	Q7	94	0	500	mV	1-5	13,14
U23	TASK ID0	58	0	600	mV	1-5	13,14
U24	TASK ID1	8					
U25	TASK ID2	60					
U26	TASK ID3	11	0	600	mV	1-5	13,14
U27	Y0	39	0	750	mV	1-5	13,14
U28	Y1	28					
U29	Y2	32					
U30	Y3	30					
U31	Y4	86					
U32	Y5	85					
U33	Y6	87					
U34	Y7	88	0	750	mV	1-5	13,14

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NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U34
Power supply current limit	0.51 A	NA	U1-U34
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U34 U1-U34
Output crosstalk measurement	 NA NA NA NA	 ± 20.0 mV ± 25.0 mV ± 30.0 mV ± 37.5 mV	 U1-U5 U6-U22 U23-U26 U27-U34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. The test circuit shall be as indicated on figure 14. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

5. Abbreviations used are defined in 6.1.

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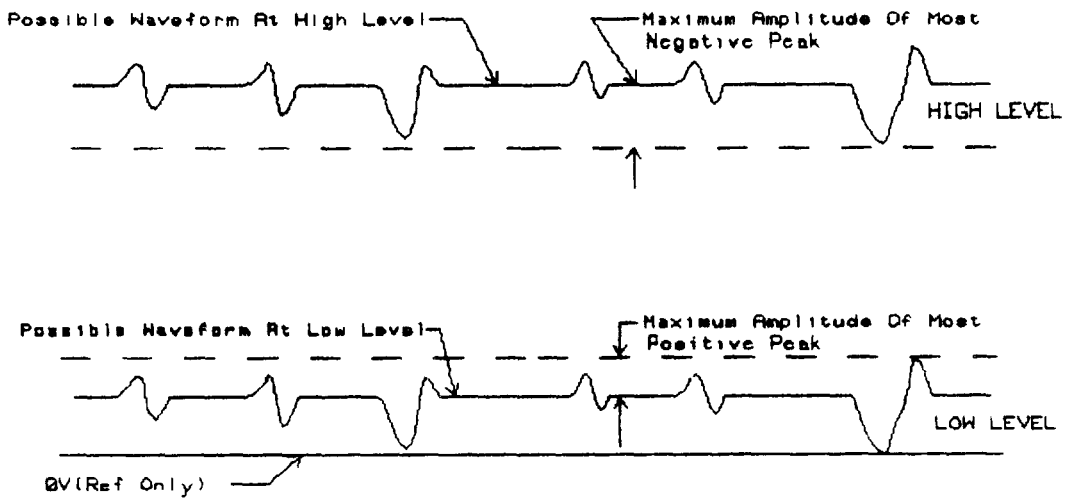


FIGURE 13. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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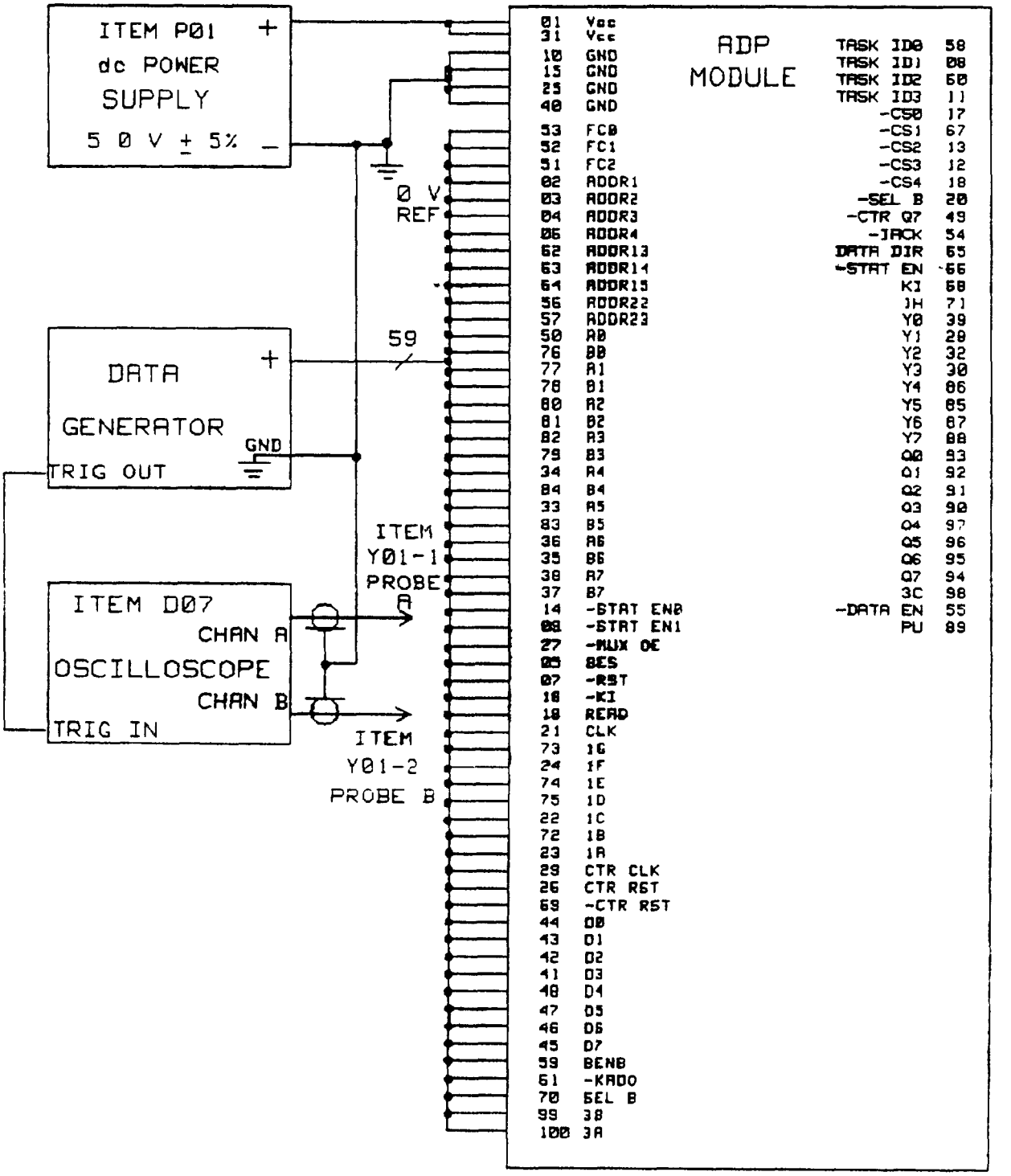


FIGURE 14. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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TABLE XXII. Output crosstalk duration, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
V1	TASK ID0	58	0	50	nS	1-4	15
V2	TASK ID1	8					
V3	TASK ID2	60					
V4	TASK ID3	11	0	50	nS	1-4	15
V5	-CS0	17	0	57	nS	1-4	15
V6	-CS1	67					
V7	-CS4	18					
V8	KI	68					
V9	DATA DIR	65					
V10	-STAT EN	66					
V11	-DATA EN	55					
V12	-CS2	13	0	57	nS	1-4	15
V13	-CS3	12	0	50	nS	1-4	15
V14	Q0	93					
V15	Q1	92					
V16	Q2	91					
V17	Q3	90					
V18	Q4	97					
V19	Q5	96					
V20	Q6	95	0	50	nS	1-4	15

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NOTES:

1. Output crosstalk duration, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	V1-V20
Power supply current limit	0.51 A	NA	V1-V20
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 V1-V20 V1-V20
Output crosstalk duration measurement	NA NA NA	± 2.5 nS ± 2.85 nS ± 2.5 nS	V1-V4 V5-V12 V13-V20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

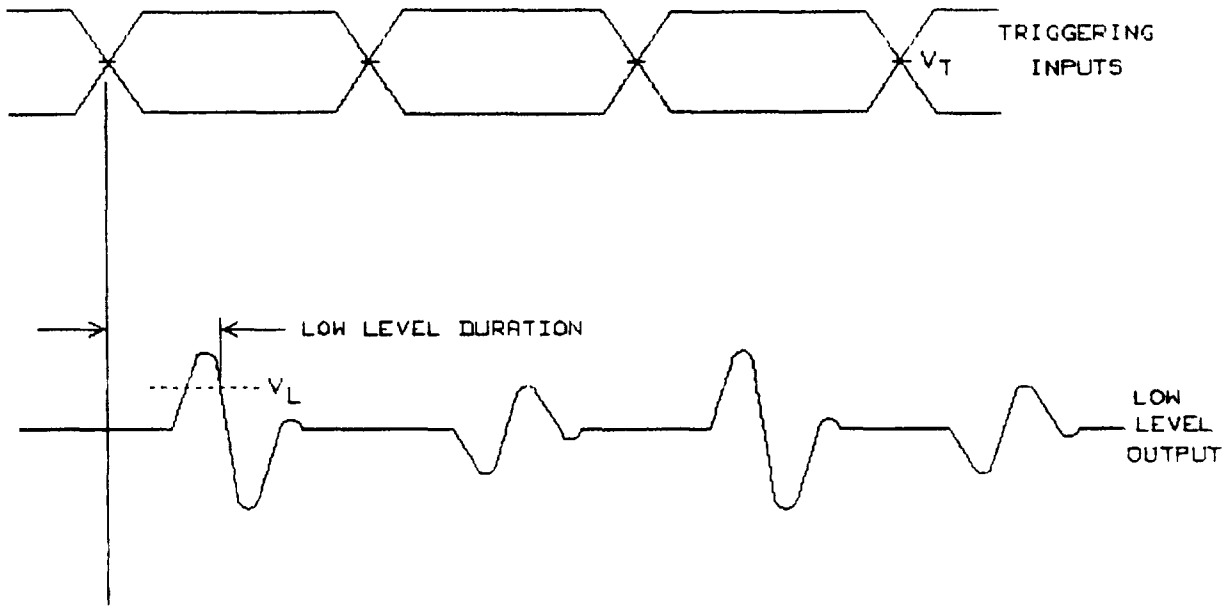
(c) The duration shall be measured from the triggering edge of the input, as shown on figure 15.

(d) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. VT = 1.3 V.

2. VL = 400 mV (ac coupled).

3. The crosstalk duration is measured from the triggering edge of the input to the point where output crosstalk returns to the specified voltage. If the crosstalk amplitude is smaller than the duration measurement voltage (VL), the crosstalk duration shall be recorded as 0 nS.

FIGURE 15. Output crosstalk duration waveforms for table XXII.

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TABLE XXIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	ADDR1	2	2	20	pF	1-3
W2	ADDR2	3				
W3	ADDR3	4	2	20	pF	1-3
W4	ADDR4	6	2	25	pF	1-3
W5	A0	50	2	20	pF	1-3
W6	A1	77				
W7	A2	80				
W8	A3	82				
W9	A4	34				
W10	A5	33	2	20	pF	1-3
W11	A6	36	2	25	pF	1-3
W12	A7	38	2	20	pF	1-3
W13	B0	76				
W14	B1	78				
W15	B2	81				
W16	B3	79				
W17	B4	84				
W18	B5	83				
W19	B6	35				
W20	B7	37	2	20	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W21	-KADO	61	2	25	pF	1-3
W22	ADDR23	57	2	35	pF	1-3
W23	ADDR22	56	2	40	pF	1-3
W24	ADDR13	62	6	50	pF	1-3
W25	ADDR14	63	4	30	pF	1-3
W26	ADDR15	64	4	30	pF	1-3
W27	BES	5	4	35	pF	1-3
W28	-STAT EN0	14	4	40	pF	1-3
W29	-STAT EN1	9	4	35	pF	1-3
W30	READ	19	2	30	pF	1-3
W31	-KI	16	2	35	pF	1-3
W32	-RST	7	2	35	pF	1-3
W33	CLK	21	2	30	pF	1-3
W34	D0	44	2	25	pF	1-3
W35	D1	43	2	25	pF	1-3
W36	D2	42	4	50	pF	1-3
W37	D3	41	6	40	pF	1-3
W38	D4	48	2	15	pF	1-3
W39	D5	47	2	20	pF	1-3
W40	D6	46	2	20	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W41	D7	45	2	15	pF	1-3
W42	CTR CLK	29				
W43	CTR RST	26	2	15	pF	1-3
W44	-CTR RST	69	2	20	pF	1-3
W45	1A	23	2	20	pF	1-3
W46	1B	72	2	15	pF	1-3
W47	1C	22				
W48	1D	75	2	15	pF	1-3
W49	1E	74	2	20	pF	1-3
W50	1F	24				
W51	1G	73				
W52	3A	100				
W53	3B	99				
W54	FC0	53				
W55	FC1	52				
W56	FC2	51				
W57	BENB	59				
W58	-MUX OE	27				
W59	SEL B	70	2	20	pF	1-3

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W59
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W59
Capacitance measurement	NA	± 0.9 pF	W1-W3
	NA	± 1.15 pF	W4
	NA	± 0.9 pF	W5-W10
	NA	± 1.15 pF	W11
	NA	± 0.9 pF	W12-W20
	NA	± 1.15 pF	W21
	NA	± 1.65 pF	W22
	NA	± 1.9 pF	W23
	NA	± 2.2 pF	W24
	NA	± 1.3 pF	W25,W26
	NA	± 1.55 pF	W27
	NA	± 1.8 pF	W28
	NA	± 1.55 pF	W29
	NA	± 1.4 pF	W30
	NA	± 1.65 pF	W31,W32
	NA	± 1.4 pF	W33
	NA	± 1.15 pF	W34,W35
	NA	± 2.3 pF	W36
	NA	± 1.7 pF	W37
	NA	± 0.65 pF	W38
	NA	± 0.9 pF	W39,W40
	NA	± 0.65 pF	W41-W43
	NA	± 0.9 pF	W44,W45
	NA	± 0.65 pF	W46-W48
	NA	± 0.9 pF	W49-W59

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X1	Y1	28	2	20	pF	1-3
X2	Y3	30				
X3	Y2	32				
X4	Y0	39				
X5	Y5	85				
X6	Y4	86				
X7	Y6	87				
X8	Y7	88	2	20	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X8
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X8
Capacitance measurement	NA	± 0.9 pF	X1-X8

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIV.

3. Abbreviations used are defined in 6.1.

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TABLE XXV. Disable time test

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Low to high impedance (tplz)								
Y1	-MUX OE to Y0	50,77,80, 82,34,33, 36,38,70	76,78,81, 79,84,83, 35,37	39	5	45	nS	16,17
Y2	-MUX OE to Y1			28				
Y3	-MUX OE to Y2			32				
Y4	-MUX OE to Y3			30				
Y5	-MUX OE to Y4			86				
Y6	-MUX OE to Y5			85				
Y7	-MUX OE to Y6			87				
Y8	-MUX OE to Y7	50,77,80, 82,34,33, 36,38,70	76,78,81, 79,84,83, 35,37	88	5	45	nS	16,17
High to high impedance (tphz)								
Y9	-MUX OE to Y0	50,77,80, 82,34,33, 36,38	76,78,81, 79,84,83, 35,37,70	39	5	66	nS	16,17
Y10	-MUX OE to Y1			28				
Y11	-MUX OE to Y2	50,77,80, 82,34,33, 36,38	76,78,81, 79,84,83, 35,37,70	32	5	66	nS	16,17

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TABLE XXV. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs.
Y12	-MUX OE to Y3	50,77,80, 82,34,33, 36,38	76,78,81, 79,84,83, 35,37,70	30	5	66	nS	16,17
Y13	-MUX OE to Y4			86				
Y14	-MUX OE to Y5			85				
Y15	-MUX OE to Y6			87				
Y16	-MUX OE to Y7	50,77,80, 82,34,33, 36,38	76,78,81, 79,84,83, 35,37,70	88	5	66	nS	16,17

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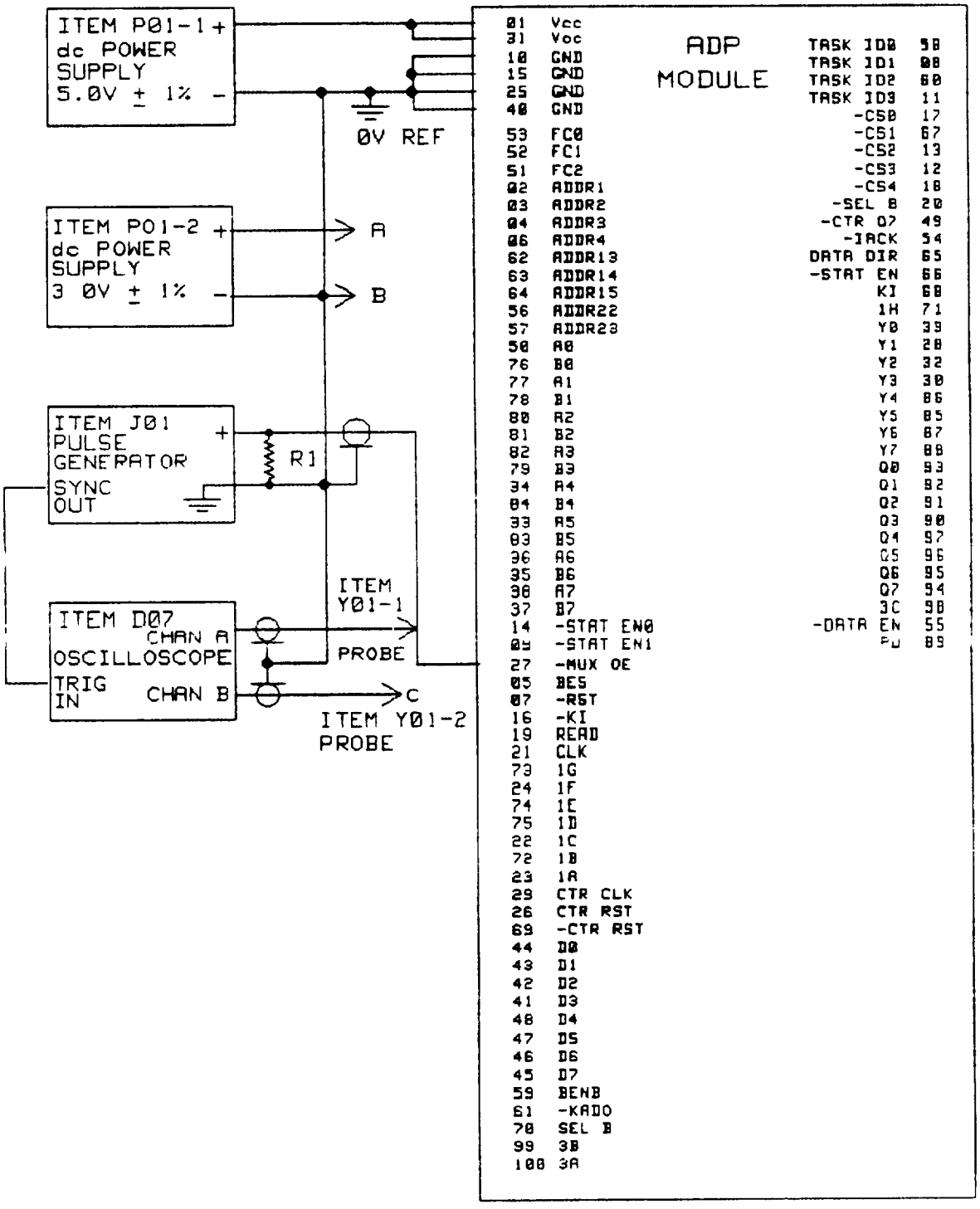


FIGURE 16. Disable time test circuit for table XXV.

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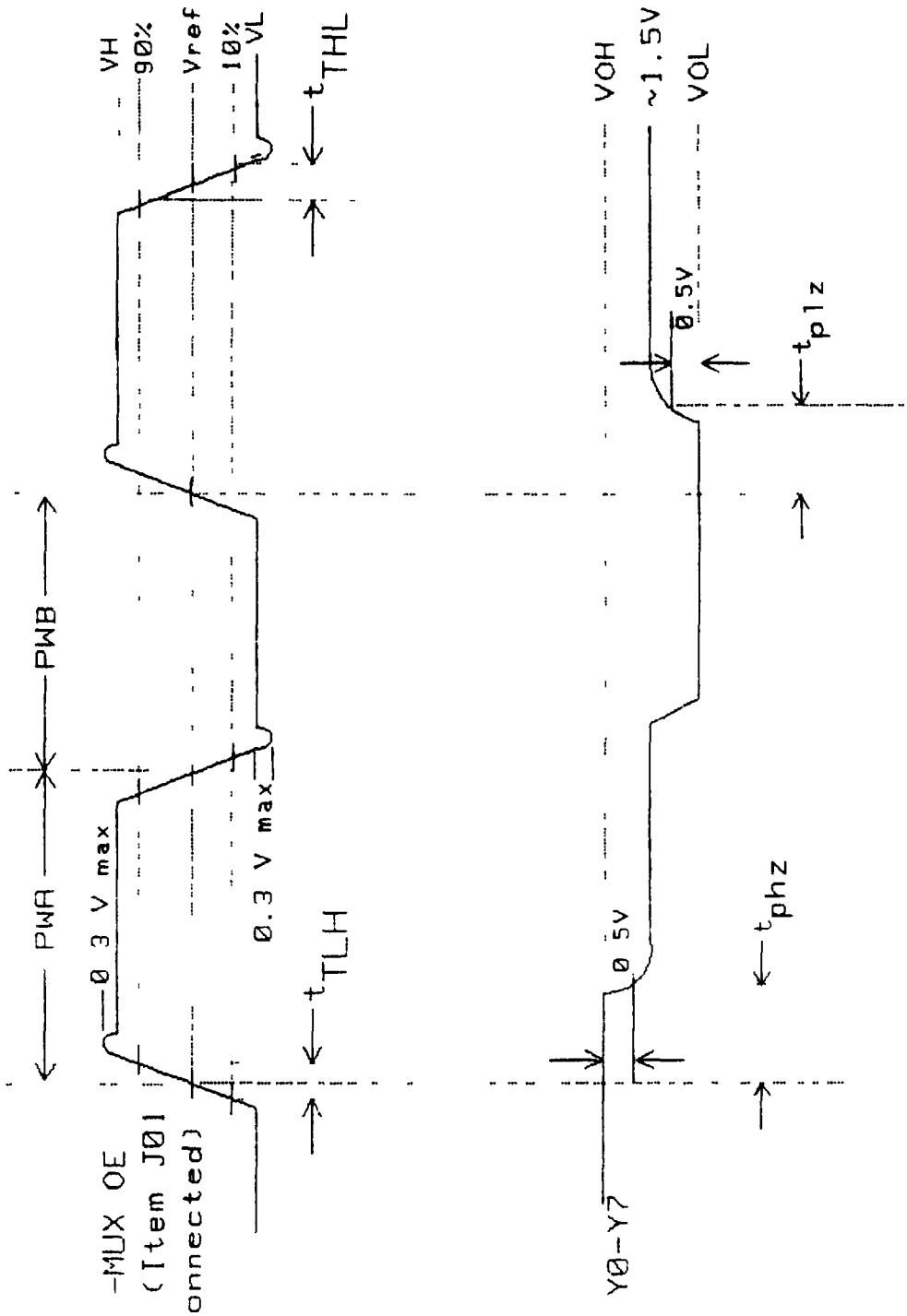


FIGURE 17. Disable time test waveforms for table XXV.

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NOTES:

1. tTLH = tTHL = 15 nS.
2. PWA = PWB = 250 nS.
3. VH = 3.0 ± 0.1 V.
4. VL = 0.0 ± 0.1 V.
5. Vref = 1.5 ± 0.1 V.

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TABLE XXVI. Threshold test.

Test cond	Test description	Notes	Fig
Z	Threshold test	1-3	9

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Z
Power supply current limit	0.51 A	NA	Z
Pattern voltage high, (VH) low, (VL) low, (VL) (see note 2)	 2.0 V 0.7 V 0.8 V	 ± 20 mV ± 7 mV ± 8 mV	 Z Z Z
Output voltage measurement high (Voh) low (Vol)	 NA NA	 ± 105 mV ± 25 mV	 Z Z

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XXVI using only a capacitive load of 100 pF (see figure 8). The output compare levels shall be as follows:

High level: 2.4 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.5 V

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

3. Abbreviations used are defined in 6.1.

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